

ABREVIATURAS Y SÍMBOLOS DE MECÁNICA

LCE	AIR ELIMINATOR ELIMINADORA DE AIRE	LCV	LEVEL CONTROL VALVE VÁLVULA DE CONTROL DE NIVEL
ATG	AUTOMATIC TANK GAUGE SENSOR AUTOMÁTICO DEL TANQUE	LPD	LOW POINT DRAIN DRENAJE PUNTO BAJO
BLDG	BUILDING EDIFICIO	M	METER METRO
BPCV	BACK PRESSURE CONTROL VALVE VÁLVULA DE CONTROL DE CONTRAPRESIÓN	mm	MILLIMETER MILÍMETRO
CONC	CONCRETE OR CONCENTRIC HORMIGÓN O CONCÉNTRICO	MIN	MINIMUM MÍNIMO
CS	CARBON STEEL ACERO AL CARBONO	NO	NUMBER NÚMERO
CV	CHECK VALVE OR CONTROL VALVE VÁLVULA ANTIRRETORNO O VÁLVULA DE CONTROL	OW	OILY WASTE RESIDUOS ACEITOSOS
DBB	DOUBLE BLOCK AND BLEED DOBLE BLOQUEO Y PURGA	PI	PRESSURE INDICATOR INDICADOR DE PRESIÓN
DEMO	DEMOLITION OR DEMOLISH DEMOLICIÓN O DEMOLER	PRT	PRODUCT RECOVERY TANK TANQUE DE RECUPERACIÓN DE PRODUCTO
DRN	DRAIN DRENAJE		
ECC	ECCENTRIC EXCÉNTRICO	RED	REDUCER REDUCTOR
FCV	FLOW CONTROL VALVE VÁLVULA DE CONTROL DE CAUDAL	SCH	SCHEDULE CUADRO, SCHEDULE
FOB	FLAT ON BOTTOM PLANO SOBRE BASE	SDD	SAFETY DEVICE AGAINST DETONATION, DRY TYPE DISPOSITIVO DE SEGURIDAD CONTRA DETONACIÓN, TIPO SECO
FL	FUSIBLE LINK ENLACE FUSIBLE	SDL	SAFETY DEVICE AGAINST DETONATION, LIQUID TYPE DISPOSITIVO DE SEGURIDAD CONTRA DETONACIÓN, TIPO LÍQUIDO
FPBV	FULL PORT BALL VALVE VÁLVULA DE BOLA DE PUERTO COMPLETO	SFI	SIGHT FLOW INDICATOR INDICADOR VISUAL DE CAUDAL
FSCV	FILTER SEPARATOR CONTROL VALVE VÁLVULA DE CONTROL DEL S EPARADOR DE FILTRO	SP	STRIPPING PUMP BOMBA DE EXTRACCIÓN
FQI	FLOW QUANTITY INDICATOR INDICADOR DE CANTIDAD DE CAUDAL	SS	STAINLESS STEEL ACERO INOXIDABLE
FS	FLOW SWITCH INTERRUPTOR DE CAUDAL	STA	STATION ESTACIÓN
FSI	FILTER/SEPARATOR (ISSUE) FILTRO/SEPARADOR (EMISOR)	TRV	THERMAL RELIEF VALVE VÁLVULA DE ALIVIO TÉRMICO
FSP	FUEL SAMPLE POINT PUNTO DE MUESTRA DE COMBUSTIBLE	TYP	TYPICAL TÍPICO
FSII	FUEL SYSTEM ICING INHIBITOR INHIBIDOR DE HIELO DEL SISTEMA DE COMBUSTIBLE	UON	UNLESS OTHERWISE NOTED A MENOS QUE SE INDIQUE OTRA COSA
FSR	FILTER/SEPARATOR (RECEIPT) FILTRO/SEPARADOR (RECEPTOR)	W/	WITH CON
HP	HORSEPOWER OR HEAT PUMP CABALLO DE VAPOR O BOMBA DE CALOR	WNF	WELD NECK FLANGE BRAZON CUELLO DE SOLDADURA
HPV	HIGH POINT VENT VENTEO PUNTO ALTO	Ø	DIAMETER PLETINA
IF	INSULATING FLANGE BRIDA AISLANTE	PL	PLATE PORCENTAJE
IV	ISSUE VENTURI EMISOR VENTURI	%	PERCENT NÚMERO
JP5	JP-5 JET FUEL SURTIDOR DE COMBUSTIBLE JP-5	#	NUMBER
kPa	KILO PASCAL KILOPASCAL		
LA	LIQUID PROBE SONDA DE LÍQUIDOS		

LEYENDA DE MECÁNICA

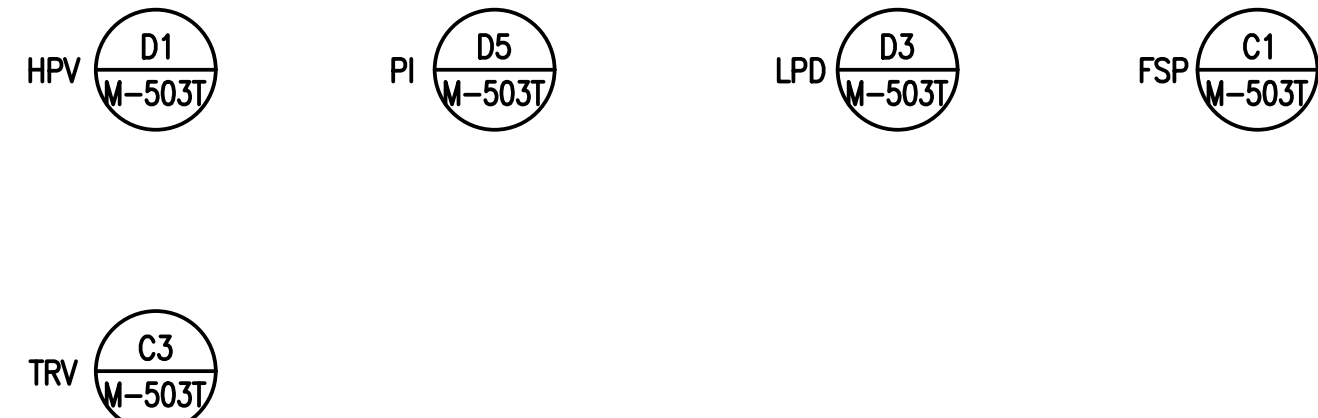
SYMBOL SÍMBOLO	DESCRIPTION DESCRIPCIÓN	
	FLOW OR PRESSURE CONTROL VALVE VÁLVULA DE CONTROL DE CAUDAL O DE PRESIÓN	
	DOUBLE BLOCK AND BLEED PLUG VALVE VÁLVULA DE DOBLE BLOQUEO Y PURGA	
	FLOW SWITCH INTERRUPTOR DE CAUDAL	
	OR	THERMAL OR SAFETY RELIEF VALVE VÁLVULA DE ALIVIO TÉRMICO O DE SEGURIDAD
	BALL VALVE VÁLVULA DE BOLA	
	BUTTERFLY VALVE VÁLVULA DE MARIPOSA	
	OR	WELD NECK FLANGE VÁLVULA CON CUELLO DE SOLDADURA
	OR	CONCENTRIC REDUCER REDUCTOR CONCENTRICO
	PRESSURE INDICATOR INDICADOR DE PRESIÓN	
	PUMP (TRIANGLE INDICATES FLOW DIRECTION) BOMBA (EL TRIÁNGULO DE LA BOMBA INDICA LA DIRECCIÓN DEL FLUJO)	
	MALE CAM TYPE DISCONNECT WITH FEMALE DUST CAP DESCONECTOR MACHO TIPO LEVA CON GUARDAPOLVO HEMBRA	
	CONNECT TO EXISTING CONECTAR AL EXISTENTE	
	MOTOR ACTUATOR ACTUADOR DEL MOTOR	
	AIR ELIMINATOR ELIMINADOR DE AIRE	
	JP5-250-AL UNDERGROUND LINE LÍNEA ENTERRADA	
	JP5-250-AL ABOVEGROUND LINE LÍNEA SOBRE EL TERRENO	
	PIPE UTILITY LINE ABBREVIATION ABREVIADO DE LÍNEA DE TUBERÍA	
	SIZE TAMAÑO	
	MATERIAL TYPE TIPO DE MATERIAL	
	BITUMINOUS CONCRETE (PLAN) HORMIGÓN BITUMINOSO (PLANTA)	
	CONCRETE (PLAN) HORMIGÓN	
	CONCRETE (SECTION) HORMIGÓN (SECCIÓN)	

NOTAS DE DEMOLICIÓN

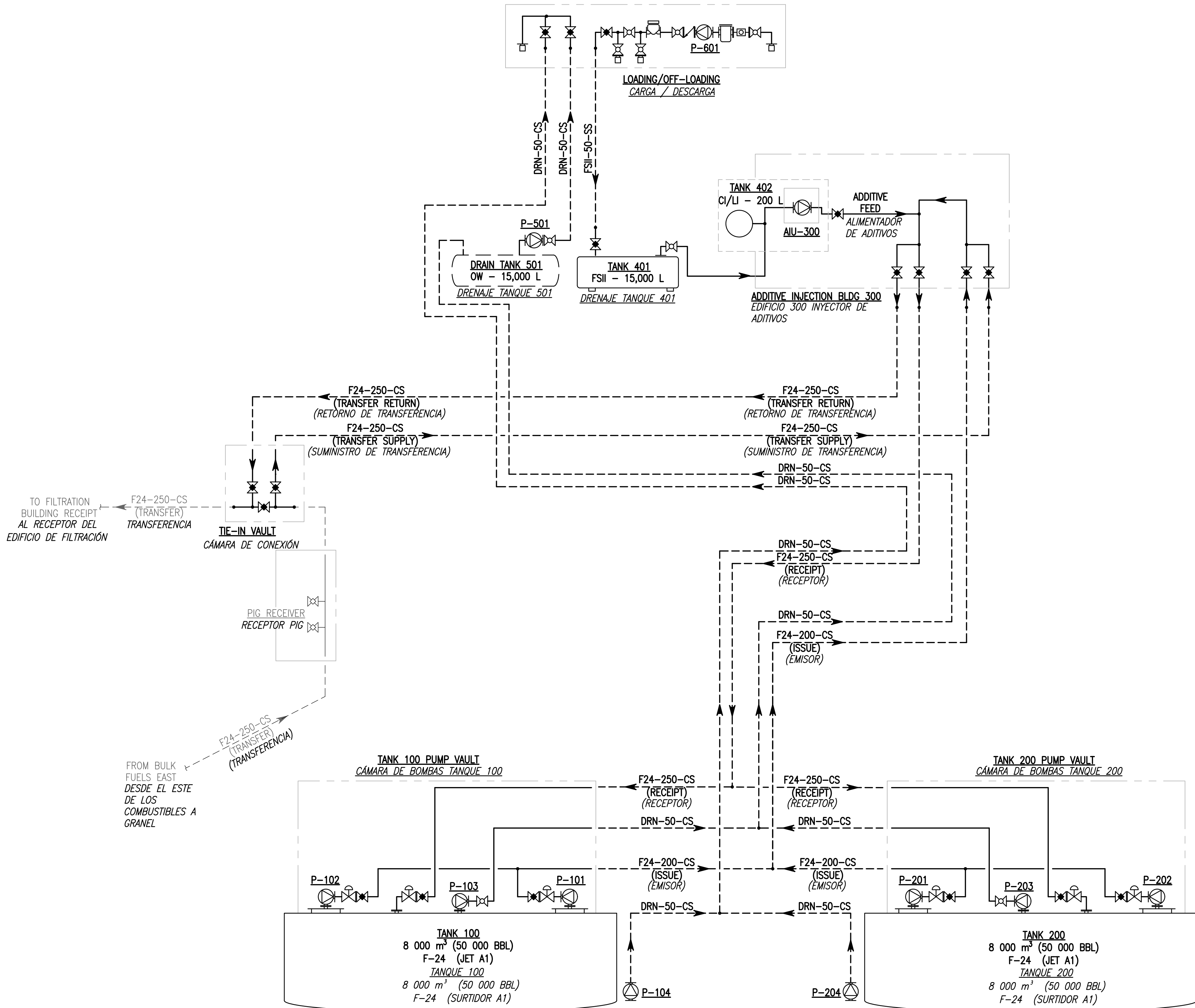
1. EXISTING FUEL SYSTEM DEMOLITION IS VERY LIMITED FOR THIS PROJECT. THE ONLY FUEL DEMOLITION REQUIRED IS ON THE INCOMING 300 mm RECEIPT PIPELINE TO ACCOMMODATE THE NEW TIE-IN VALVE VAULT. ALL OTHER ASPECTS OF THIS PROJECT ARE NEW AND DO NOT REQUIRE DEMOLITION OF EXISTING FUEL SYSTEM COMPONENTS.
1. LA DEMOLICION DEL SISTEMA DE COMBUSTIBLE EXISTENTE ES MUY LIMITADA EN ESTE PROYECTO. LA ÚNICA DEMOLICIÓN DE COMBUSTIBLE REQUERIDA ESTÁ EN LA ENTRADA DE LA TUBERÍA RECEPTORA DE 300mm PARA ADAPTARSE A LA NUEVA CÁMARA DE CONEXIÓN DE VÁLVULA. TODOS LOS OTROS ASPECTOS EN ESTE PROYECTO SON NUEVOS Y NO REQUIEREN LA DEMOLICIÓN DE COMPONENTES DEL SISTEMA DE COMBUSTIBLE EXISTENTE.
2. UNLESS OTHERWISE INDICATED, ALL EXISTING FUEL PIPING IS EXTERIOR COATED CARBON STEEL.
2. A MENOS QUE SE INDIQUE OTRA COSA, TODAS LAS TUBERÍAS DE COMBUSTIBLE EXISTENTES SON CON REVESTIMIENTO EXTERIOR DE ACERO AL CARBONO.
3. "DEMOLISH" - REMOVE IN ITS ENTIRETY.
3. "DEMOLISH" - RETIRAR COMPLETAMENTE.
4. "ABANDON IN PLACE" - CLEAN PIPING BY PIGGING, FILL COMPLETELY WITH GROUT, AND CAP OPEN ENDS OF PIPE WITH A WELDED PIPE CAP OR BLIND FLANGE. UON, WHEN ABANDONING IN PLACE PIPING WITH ABOVEGROUND PORTIONS, CUT PIPING 600 mm BELOW GRADE BEFORE PROCEEDING AS INDICATED ABOVE.
4. ABANDONAR EN PLAZA - LIMPIAR LA TUBERÍA MEDIANTE PIGGING, RELLENAR COMPLETAMENTE CON MORTERO Y TAPONAR LOS EXTREMOS ABIERTOS DE LA TUBERÍA CON UNA TAPA DE TUBERÍA SOLDADA O UNA BRIDA CIEGA. A MENOS QUE SE INDIQUE OTRA COSA, CUANDO SE ABANDONE IN SITU UNA TUBERÍA CON PARTES POR ENCIMA DEL SUELO, CORTAR LA TUBERÍA 600 mm POR DEBAJO DEL TERRENO ANTES DE PROCEDER COMO SE HA INDICADO ANTERIORMENTE.
5. THE ACTIVITY WILL, TO THE FULLEST EXTENT POSSIBLE, EMPTY THE ACTIVE UNDERGROUND AND ABOVEGROUND JP-5 PIPELINES, TANKS AND EQUIPMENT. HOWEVER, FOR BASIS OF BIDS THE PIPES ARE FULL OF FUEL ON TOP OF 50 mm OF PETROLEUM CONTAMINATED SLUDGE AND WATER, AND THE TANKS ARE FULL OF FUEL ON TOP OF 150 mm OF PETROLEUM CONTAMINATED SLUDGE AND WATER. THE CONTRACTOR MUST DRAIN, COLLECT, AND TURN OVER TO THE GOVERNMENT ANY FUEL NOT EMPTIED BY THE ACTIVITY.
5. LA ACTIVIDAD VACIARÁ, EN LA MAYOR MEDIDA POSIBLE, LAS TUBERÍAS, DEPÓSITOS Y EQUIPOS SUBTERRÁNEOS Y AÉREOS ACTIVOS DEL JP-5. SIN EMBARGO, PARA LA OFERTA BASE LAS TUBERÍAS ESTÁN LLENAS DE COMBUSTIBLE POR ENCIMA DE 50 mm DE LODOS CONTAMINADOS CON PETRÓLEO Y AGUA, Y LOS DEPÓSITOS ESTÁN LLENOS DE COMBUSTIBLE POR ENCIMA DE 150 mm DE LODOS CONTAMINADOS CON PETRÓLEO Y AGUA. EL CONTRATISTA VACIARÁ, RECOGERÁ Y ENTREGARÁ AL GOBIERNO TODO EL COMBUSTIBLE QUE NO HAYA SIDO VACIADO POR LA ACTIVIDAD.

NOTAS DE MECÁNICA

1. ALL PIPING IS EXTERIOR COATED CARBON STEEL, BOTH ABOVEGROUND AND UNDERGROUND AS INDICATED, WITH THE EXCEPTION OF THE PIPING AND EQUIPMENT FOR THE FUEL ADDITIVE SYSTEM OR AS OTHERWISE NOTED. FUEL ADDITIVE SYSTEM PIPING AND EQUIPMENT IS STAINLESS STEEL, BOTH ABOVEGROUND AND UNDERGROUND AS INDICATED.
1. LAS TUBERÍA DE COMBUSTIBLES EN ESTE PROYECTO ESTARÁN REVESTIDAS CON ACERO AL CARBONO, TANTO AÉREAS COMO ENTERRADAS SEGÚN SE INDIQUE, CON LA EXCEPCIÓN DE LA TUBERÍA Y EQUIPO PARA EL SISTEMA DE ADITIVOS DEL COMBUSTIBLE O SI SE INDICA LO CONTRARIO. LAS TUBERÍAS DEL SISTEMA DE ADITIVOS DEL COMBUSTIBLE Y LOS EQUIPOS SERÁN DE ACERO INOXIDABLE, TANTO AÉREAS COMO ENTERRADAS, SEGÚN SE INDIQUE.
2. PROVIDE HIGH POINT VENTS WHERE INDICATED AND AT ALL HIGH POINTS. PROVIDE LOW POINT DRAINS WHERE INDICATED AND AT ALL LOW POINTS. INSTALL FUEL PIPING WITH A MINIMUM SLOPE OF 0.2% DOWN FROM HIGH POINTS TO LOW POINTS, UNLESS OTHERWISE INDICATED.
2. PROVEER VENTOS DE PUNTOS ALTOS DONDE SE INDIQUE Y EN TODOS LOS PUNTOS ALTOS. PROVEER DRENAJES DE PUNTOS BAJOS DONDE SE INDIQUE Y EN TODOS LOS PUNTOS BAJOS. INSTALAR LA TUBERÍA DE COMBUSTIBLE CON UNA PENDIENTE MÍNIMA DEL 0.2% DESCENDENTE DESDE LOS PUNTOS ALTOS HASTA LOS PUNTOS BAJOS, A MENOS QUE SE INDIQUE OTRA COSA.
3. CONTRACTOR MUST PERFORM EXPLORATORY OPERATIONS AS NECESSARY TO VERIFY THE LOCATIONS, ELEVATIONS, AND DIMENSIONS OF ALL EXISTING UNDERGROUND UTILITIES AND OBSTRUCTIONS. HAND EXCAVATE WITHIN 1 METER OF EXISTING UTILITIES.
3. EL CONTRATISTA REALIZARÁ LAS OPERACIONES DE EXPLORACIÓN NECESARIAS PARA VERIFICAR LAS POSICIONES, ELEVACIONES Y DIMENSIONES DE TODAS LAS OBSTRUCCIONES ENTERRADAS. EXCAVAR MANUALMENTE HASTA 1 METRO DE LAS INSTALACIONES EXISTENTES.
4. EXISTING FUEL SYSTEMS MUST REMAIN IN OPERATION THROUGHOUT THE PROJECT EXCEPT FOR INDIVIDUAL PIPELINES AND SYSTEMS TAKEN OUT OF SERVICE TO MAKE THE FINAL PIPING CONNECTIONS.
4. LOS SISTEMAS DE COMBUSTIBLES EXISTENTES PERMANECERÁN EN FUNCIONAMIENTO A LO LARGO DEL PROYECTO EXCEPTO LAS TUBERÍAS Y SISTEMAS INDIVIDUALES QUE SE PONGAN FUERA DE SERVICIO BREVEMENTE PARA REALIZAR LAS CONEXIONES FINALES DE LAS TUBERÍAS.
5. BURIED FUEL PIPING MUST NOT BE BACK FILLED WITHOUT COMPLETING THE COATING, PNEUMATIC, WELD RADIOGRAPHY AND OTHER PIPING EXAMINATIONS AND APPROVALS. IF INSTALLED PIPE IS EMPTY WITHOUT PRODUCT FOR MORE THAN 30 DAYS, CONTRACTOR MUST CHARGE INACTIVE PIPELINES WITH A NITROGEN BLANKET TO PREVENT INTERNAL PIPE CORROSION. ENERGIZE AND BALANCE THE CURRENT CATHODIC PROTECTION SYSTEMS WITHIN 30 DAYS OF BACK FILLING ANY PIPING.
5. LAS TUBERÍAS DE COMBUSTIBLE ENTERRADAS NO SE VOLVERÁN A LLENAR SIN HABER COMPLETADO EL REVISADO Y APROBACIÓN DEL REVESTIMIENTO, PNEUMÁTICA, RADIOGRAFÍA DE SOLDADURAS Y OTRAS. SI LA TUBERÍA INSTALADA ESTÁ VACÍA SIN PRODUCTO DURANTE MÁS DE 30 DÍAS, EL CONTRATISTA CARGARÁ LAS TUBERÍAS INACTIVAS CON UNA MANTA DE NITRÓGENO PARA EVITAR LA CORROSIÓN INTERNA DE LA TUBERÍA. ENERGIZAR Y EQUILIBRAR LOS SISTEMAS ACTUALES DE PROTECCIÓN CÁTODICA EN UN PLAZO DE 30 DÍAS DESPUÉS DE RELLENAR CUALQUIER TUBERÍA.
6. THIS MILCON PROJECT FEATURES TWO DISTINCTLY SEPARATE FUEL SYSTEM COMPONENTS, THE F-24 STORAGE TANK AREA AND THE JP-5 CARGO PIPELINE. THERE IS NO INTERCONNECTION BETWEEN THE TWO PROJECTS, AND PIPING, EQUIPMENT, AND COMPONENTS BETWEEN THE TWO SEPARATE SYSTEMS MAY BE DIFFERENT PIPING MATERIALS, DIFFERENT COATING SYSTEMS, DIFFERENT PRESSURE CLASS, ETC. CONTRACTOR MUST BE AWARE THAT DETAILS SHOWN ON VARIOUS SHEETS AND ITEMS LISTED WITHIN THE SPECIFICATIONS MAY APPLY TO THE F-24 STORAGE TANK SYSTEM, THE JP-5 PIPELINE SYSTEM, OR BOTH (AS INDICATED OR NOTED).
6. ESTE PROYECTO MILCÓN PRESENTA DOS COMPONENTES DEL SISTEMA DE COMBUSTIBLE CLARAMENTE SEPARADOS, LA ZONA DEL TANQUE DE ALMACENAMIENTO F-24 Y LA TUBERÍA DE CARGA JP-5. NO HAY INTERCONEXIÓN ENTRE LOS DOS PROYECTOS, Y LAS TUBERÍAS, EQUIPOS Y COMPONENTES ENTRE LOS DOS SISTEMAS SEPARADOS PUEDEN SER DE MATERIALES DE TUBERÍA DIFERENTES, SISTEMAS DE REVESTIMIENTO DIFERENTES, CLASE DE PRESIÓN DIFERENTE, ETC. EL CONTRATISTA DEBE SER CONSCIENTE DE QUE LOS DETALLES MOSTRADOS EN VARIOS DE LOS PLANOS Y LOS ELEMENTOS ENUMERADOS EN LAS ESPECIFICACIONES PUEDEN APLICARSE AL SISTEMA DE TANQUES DE ALMACENAMIENTO DEL F-24, AL SISTEMA DE TUBERÍAS DEL JP-5 O A AMBOS (SEGÚN SE INDIQUE O SEÑALE).
7. HPV, LPD, TRV, PI, FSP, AND IF ARE INDICATED ON THE PLANS WITH A LEADER AND AN ABBREVIATION. THE DETAILS ARE LOCATED AS FOLLOWS:
7. HPV, LPD, TRV, PI, FSP, Y IF SE INDICAN EN LOS PLANOS CON UNA SIGLA Y UNA ABREVIATURA. LOS DETALLES SE INDICAN A CONTINUACIÓN:



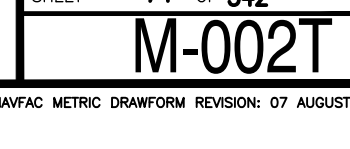
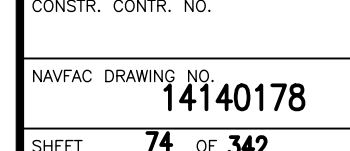
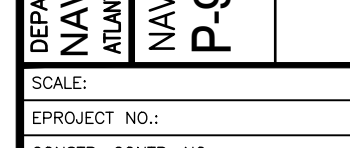
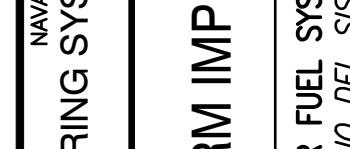
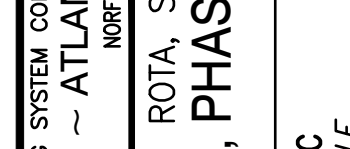
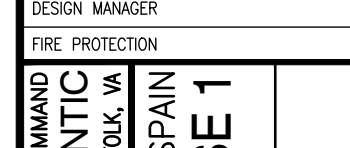
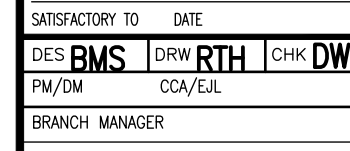
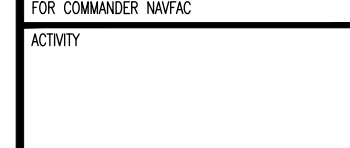
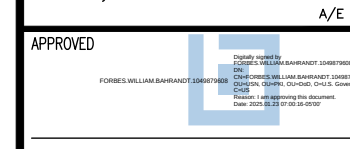
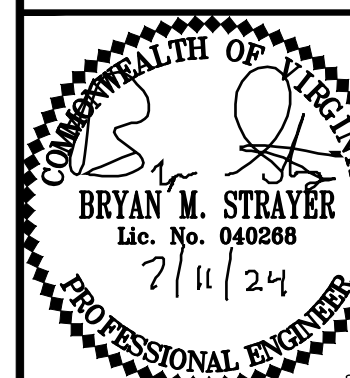
FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\NAV-002T CUT-AND-COVER FUEL SYSTEM FLOW SCHEMATIC.dwg LAYOUT NAME: m-002 PLOTTED: Tuesday, July 09, 2024 - 1:38pm USER: rhopkins



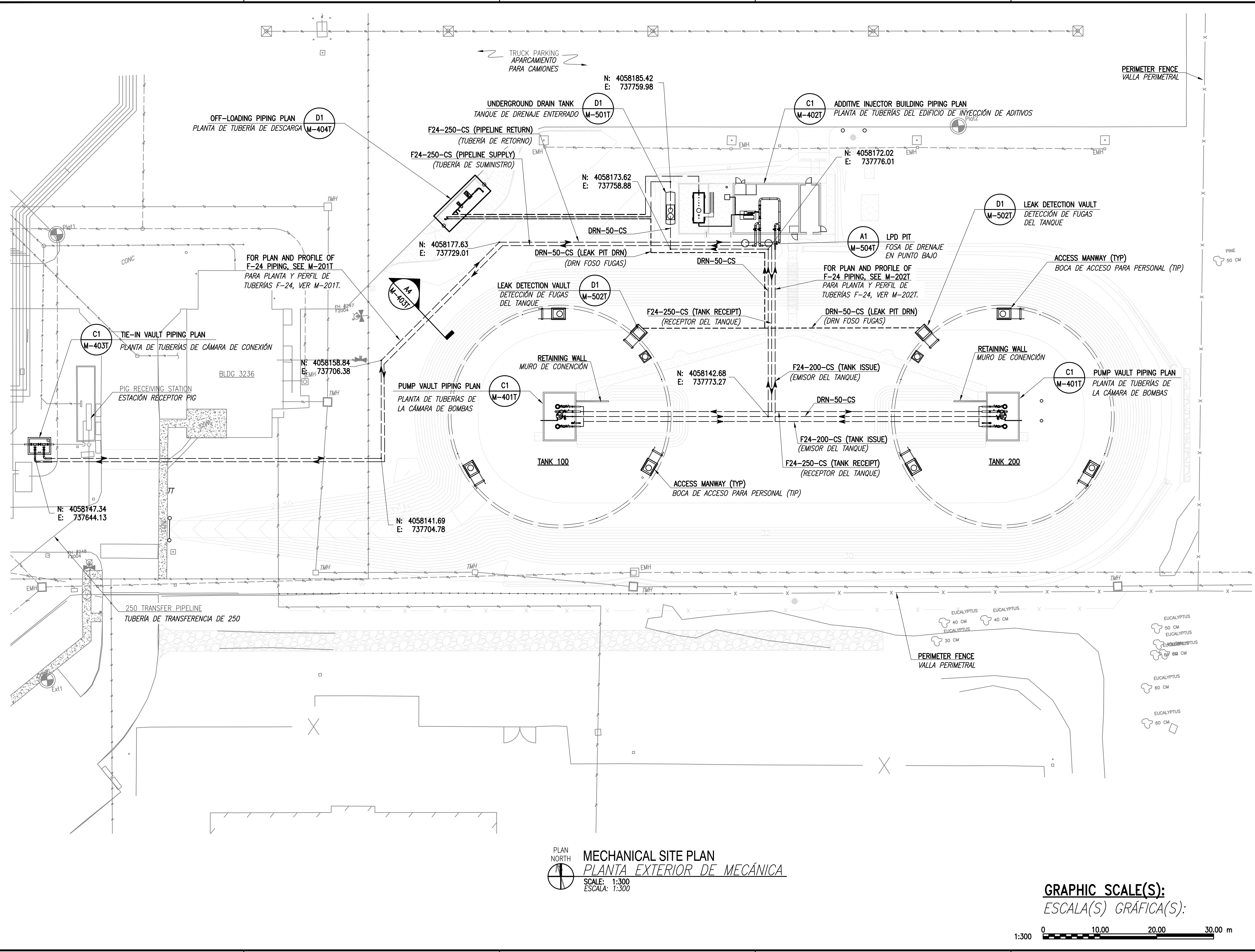
CUT-AND-COVER FUEL SYSTEM FLOW SCHEMATIC
DIAGRAMA DE FLUJO DEL SISTEMA DE COMBUSTIBLE
SCALE: NONE
ESCALA: SIN ESCALA

FUELING SYSTEM NOTES NOTAS DEL SISTEMA DE COMBUSTIBLE

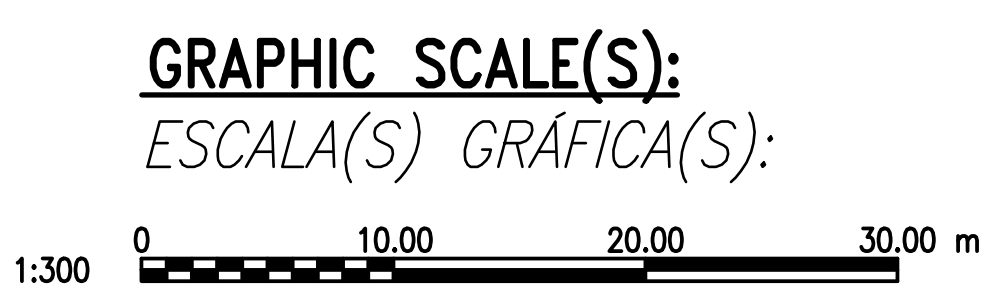
- EACH OF THE NEW BULK STORAGE TANKS (TANK 100 AND TANK 200) WILL BE PROVIDED WITH A PUMP VAULT AND (2) 600 GPM VERTICAL TURBINE PUMPS. MAXIMUM FLOW RATE WITH BOTH PUMPS RUNNING SIMULTANEOUSLY WILL BE 1,200 GPM.
CADA UNO DE LOS NUEVOS TANQUES DE ALMACENAMIENTO A GRANEL (TANQUE 100 Y TANQUE 200) SE PROVEERÁ CON UNA CÁMARA DE BOMBAS Y (2) BOMBAS VERTICALES DE TURBINA DE 600 GPM. EL CAUDAL MÁXIMO SERÁ DE 1.200 GPM CON AMBAS BOMBAS FUNCIONANDO SIMULTÁNEAMENTE.
- THE ADDITIVE INJECTION BUILDING (BLDG 300) HAS THE CAPABILITY TO ALLOW FUEL TO PASS THROUGH OR TO ADDITIVE FUEL TO ACCOMMODATE SEVERAL DIFFERENT TRANSFER SCENARIOS. ADDITIVE INJECTION SKID CAN BE ISOLATED FROM PIPING SYSTEM USING ISOLATION VALVES IN THE FEED LINE.
EL EDIFICIO DEL INYECTOR DE ADITIVOS (EDIFICIO 300) TIENE CAPACIDAD PARA PERMITIR EL PASO DEL COMBUSTIBLE A TRAVÉS DE ÉL O PARA ADITIVARLO CON EL FIN DE ADAPTARSE A VARIOS ESCENARIOS DE TRANSFERENCIA DIFERENTES. EL SISTEMA DE INYECCIÓN DE ADITIVOS PUEDE AISLARSE DEL SISTEMA DE TUBERÍAS MEDIANTE VÁLVULAS DE AISLAMIENTO EN LA LÍNEA DE ALIMENTACIÓN.
- FOR ANY OF THE ADDITIZATION SCENARIOS, THE FLOW RATE WILL BE 1,200 GPM. THE VARIOUS ADDITIZATION TRANSFER SCENARIOS ARE AS FOLLOWS:
PARA CUALQUIERA DE LOS ESCENARIOS DE ADICIÓN, EL CAUDAL SERÁ DE 1.200 GPM. LOS DISTINTOS ESCENARIOS DE TRANSFERENCIA DE ADICIÓN SON LOS SIGUIENTES:
 - FUEL VIA TRANSFER LINE IS ADDITIZED THEN RECEIVED TO EITHER OF THE NEW BULK STORAGE TANKS (TANK 100 AND TANK 200).
EL COMBUSTIBLE A TRAVÉS DE LA LÍNEA DE TRANSFERENCIA ES ADITIVADO Y ENTONCES RECIBIDO POR CUALQUIERA DE LOS NUEVOS TANQUES DE ALMACENAJE (TANQUE 100 Y TANQUE 200).
 - FUEL VIA TRANSFER LINE IS ADDITIZED THEN RECEIVED TO EITHER OF THE EXISTING HYDRANT SYSTEM OPERATING TANKS.
EL COMBUSTIBLE A TRAVÉS DE LA LÍNEA DE TRANSFERENCIA ES ADITIVADO Y ENTONCES RECIBIDO POR CUALQUIERA DE LOS TANQUES EN FUNCIONAMIENTO DEL SISTEMA DE HIDRANTES EXISTENTE.
 - FUEL WITHDRAWN FROM EITHER OF THE NEW BULK STORAGE TANK IS ADDITIZED THEN RECEIVED INTO EITHER OF THE EXISTING HYDRANT SYSTEM OPERATING TANKS.
EL COMBUSTIBLE EXTRAÍDO DE CUALQUIERA DE LOS NUEVOS TANQUES DE ALMACENAMIENTO SE ADITIVA Y LUEGO SE RECIBE EN CUALQUIERA DE LOS TANQUES EN FUNCIONAMIENTO DEL SISTEMA DE HIDRANTES EXISTENTE.
 - FUEL CAN BE WITHDRAWN FROM EITHER OF THE NEW BULK STORAGE TANKS, ADDITIZED, THEN RECEIVED INTO THE OTHER NEW BULK STORAGE TANK.
EL COMBUSTIBLE PUEDE RETIRARSE DE CUALQUIERA DE LOS NUEVOS TANQUES DE ALMACENAMIENTO A GRANEL, ADITIVARSE Y, A CONTINUACIÓN, RECIBIRSE EN EL OTRO NUEVO TANQUE DE ALMACENAMIENTO A GRANEL.



FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\A\A-M-101T MECHANICAL SITE PLAN.dwg LAYOUT NAME: M-101T PLOTTED: Tuesday, July 09, 2024 - 1:39pm USER: hopkins



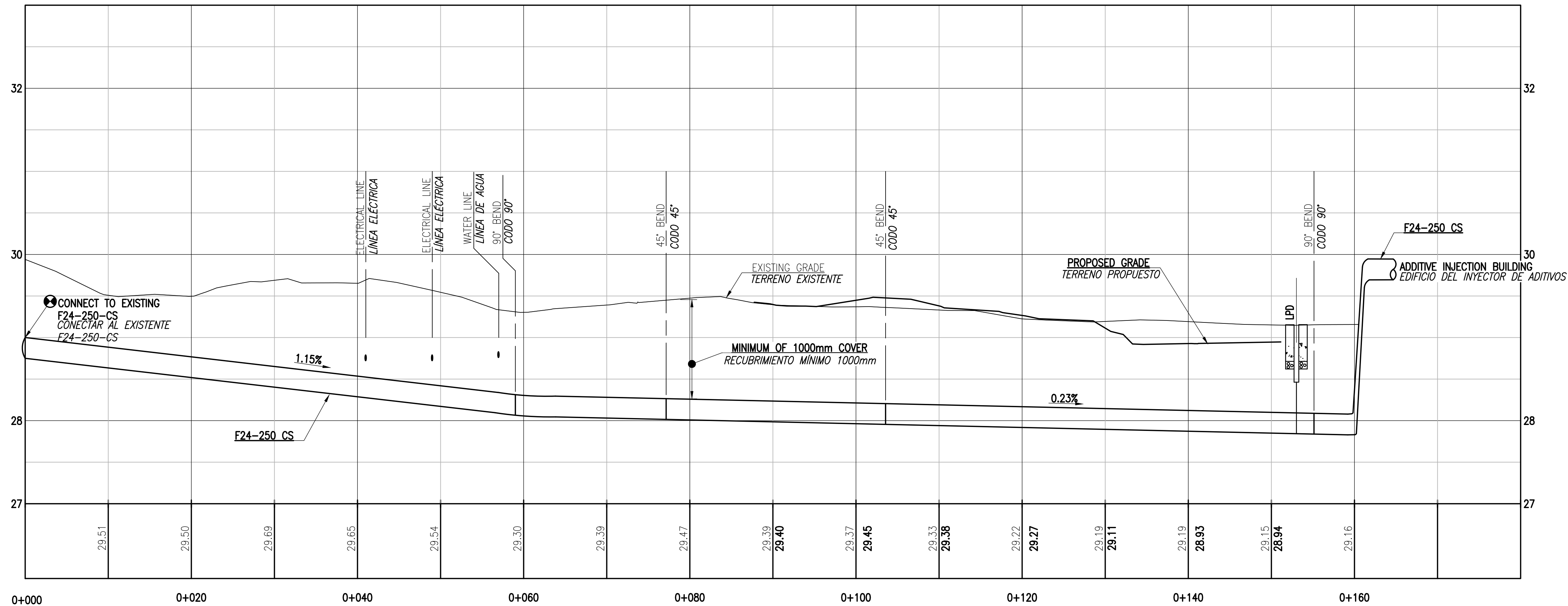
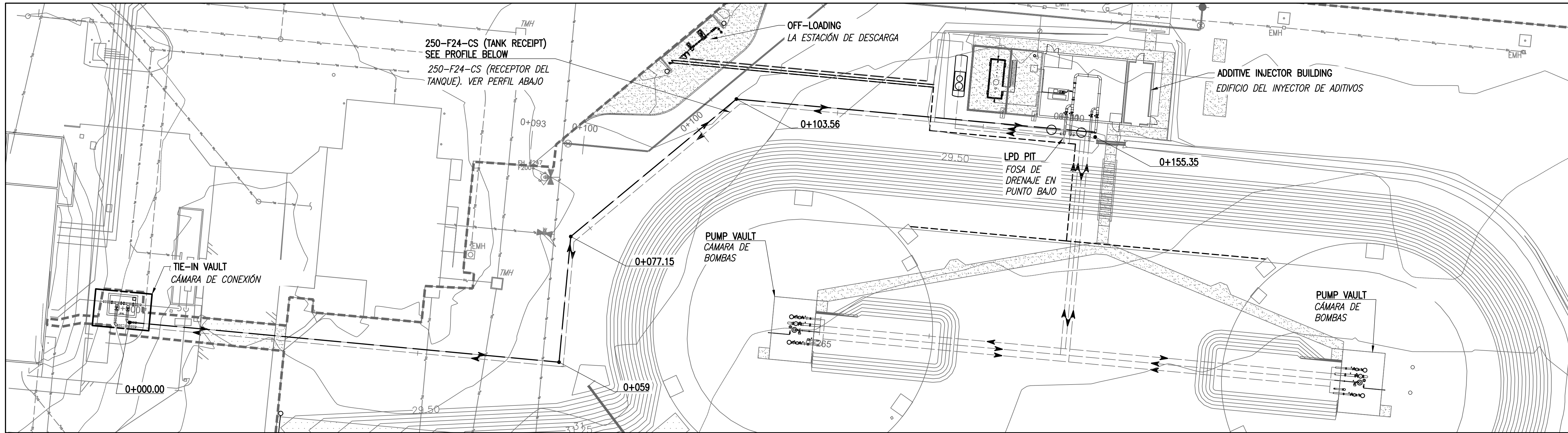
PLAN NORTH
MECHANICAL SITE PLAN
PLANTA EXTERIOR DE MECÁNICA
SCALE: 1:300
ESCALA: 1:300



APPROVED		DATE	APPR
DESCRIPTION		DATE	APPR
SYMBOL		DATE	APPR
Brockenbrough ENGINEERING - CONSULTING 1011 Boulder Springs Drive, Suite 200, Chesapeake, Virginia 23026 (804) 552-2500 (fax) (804) 552-2501 (cell) www.brockenbrough.com			
Stanley Consultants Inc. 225 Iowa Avenue, Muscatine, Iowa 52761-3764 www.stanleyconsultants.com			
APPROVED			
FOR COMMANDER NAVFAC			
ACTIVITY			
SATISFACTORY TO: DATE			
DES: BMS	DRAW: RTH	CHK: DWN	
PM/CM	CC/CA	CC/CA	
BRANCH MANAGER			
DESIGN MANAGER			
FIRE PROTECTION			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC			
ATLANTIC DESIGN AND CONSTRUCTION			
NAVAL STATION ROTA			
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1			
ROTA, SPAIN			
MECHANICAL SITE PLAN			
PLANTA EXTERIOR DE MECÁNICA			
NAVFAC DRAWING NO: 14140179			
SHEET 75 OF			
M-101T			
NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018			

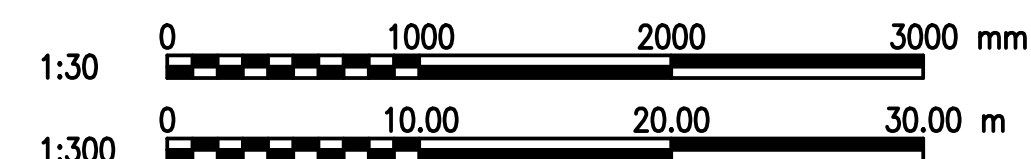
FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\M\201T FUEL PIPELINE PLANK.dwg LAYOUT NAME: m-201 PLOTTED: Tuesday, July 09, 2024 - 1:39pm USER: rhopkins

A
B
C
D



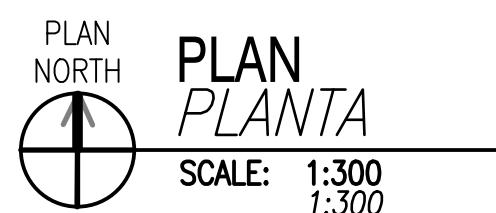
FUEL LINE PROFILE
PERFIL DE LA LÍNEA DE COMBUSTIBLE
SCALE: 1:300 HORIZONTAL
1:30 VERTICAL

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):



DATE	APPR
DESCRIPTION	SYN
Brockenbrough ENGINEERING & CONSTRUCTION 1011 Boulder Springs Drive, Suite 200, Chesapeake, Virginia 23060 (804) 552-2800 (fax) (804) 552-2801 (cell) www.brockenbrough.com	
Stanley Consultants Inc. 225 Iowa Avenue, Muscatine, Iowa 52761-3764 www.stanleyconsultants.com A/E INFO	
APPROVED 	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO: DATE	
DESIGNER: BMS	DRAWN: RTH
IN/OUT: CCA/CLA	CHK: DWN
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC ATLANTIC DESIGN AND CONSTRUCTION	
NAVAL STATION ROTA P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
ROTA, SPAIN	
FUEL PIPELINE PLAN PLANTA DE TUBERÍAS DE COMBUSTIBLE	
SCALE: EPROJECT NO.: CONSTR. CONTR. NO.	
NAVFAC DRAWING NO: 14140180	
SHEET 76 OF 342	
M-201T	
NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018	

A
B
C
D



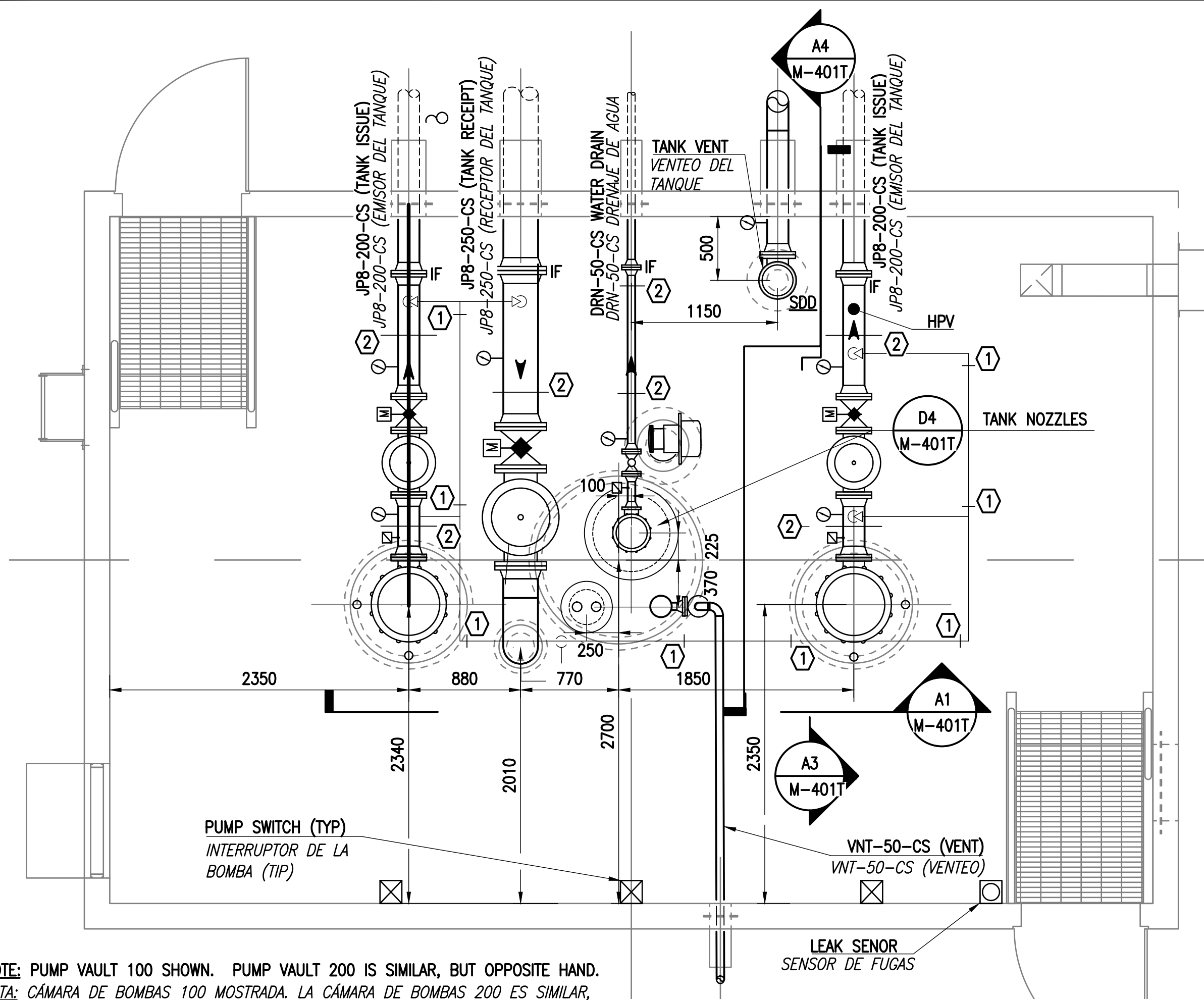
FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Roto\CAD\M-M-401T TANK VAULT PIPING PLAN.dwg LAYOUT NAME: M-401 PLOTTED: Tuesday, July 09, 2024 - 1:40pm USER: mhopkins

D

C

B

A



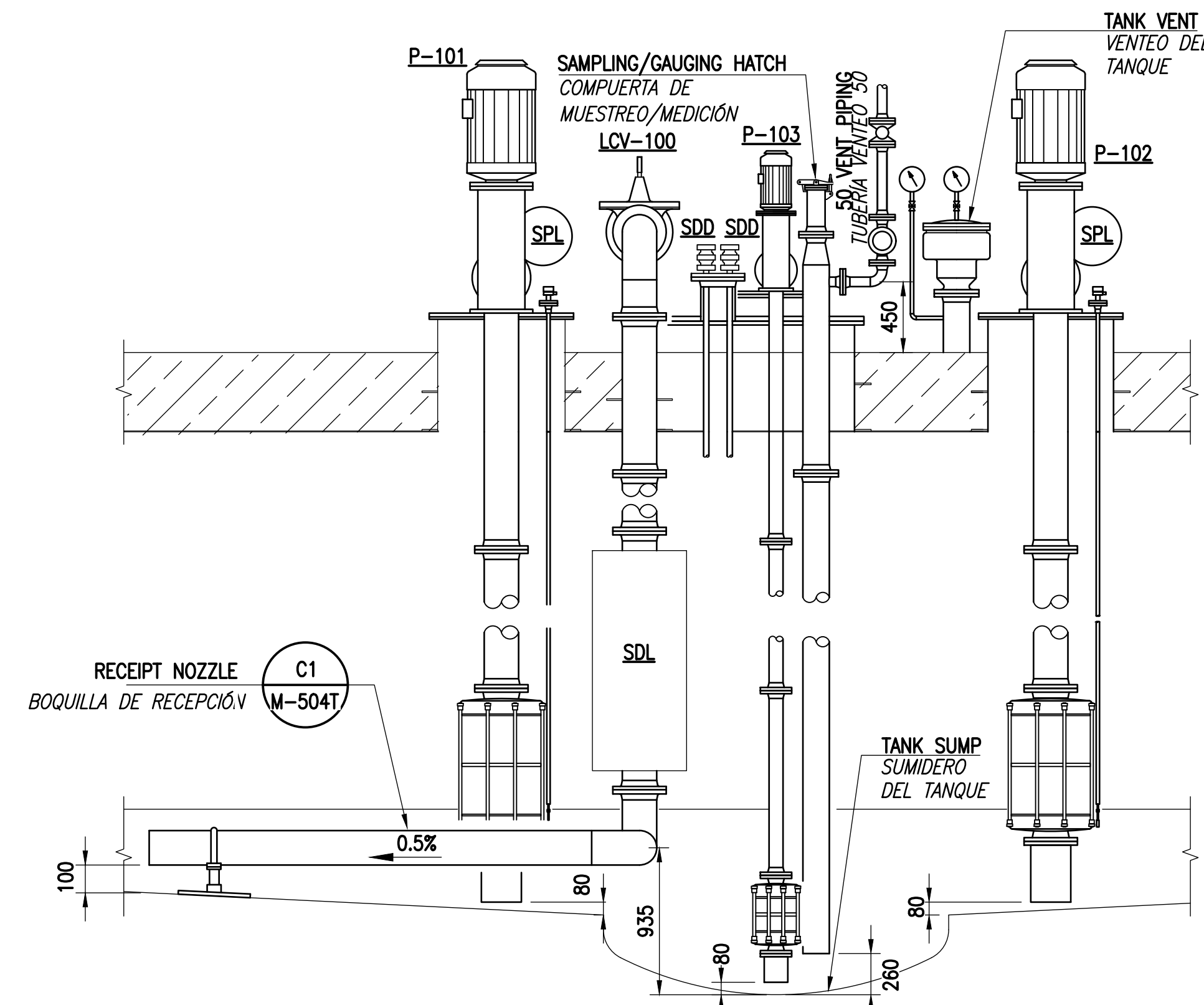
NOTE: PUMP VAULT 100 SHOWN. PUMP VAULT 200 IS SIMILAR, BUT OPPOSITE HAND.
NOTA: CÁMARA DE BOMBAS 100 MOSTRADA. LA CÁMARA DE BOMBAS 200 ES SIMILAR, PERO SIMÉTRICA

PUMP VAULT PIPING PLAN

PLANTA DE TUBERÍAS DE LA CÁMARA DE BOMBAS

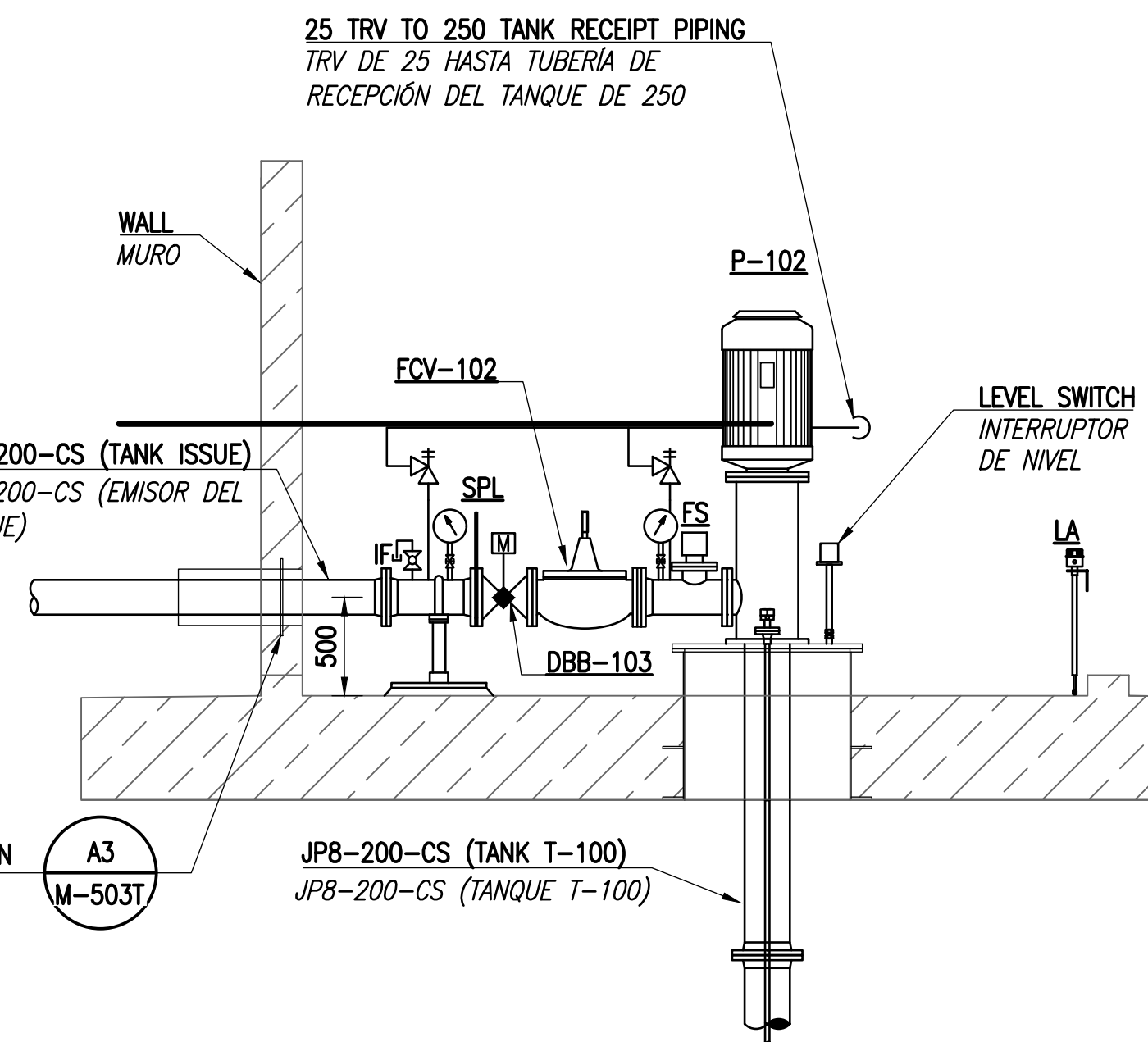
C1

SCALE: 1:30
ESCALA: 1:30



SECTION
SECCIÓN
SCALE: 1:30
ESCALA: 1:30

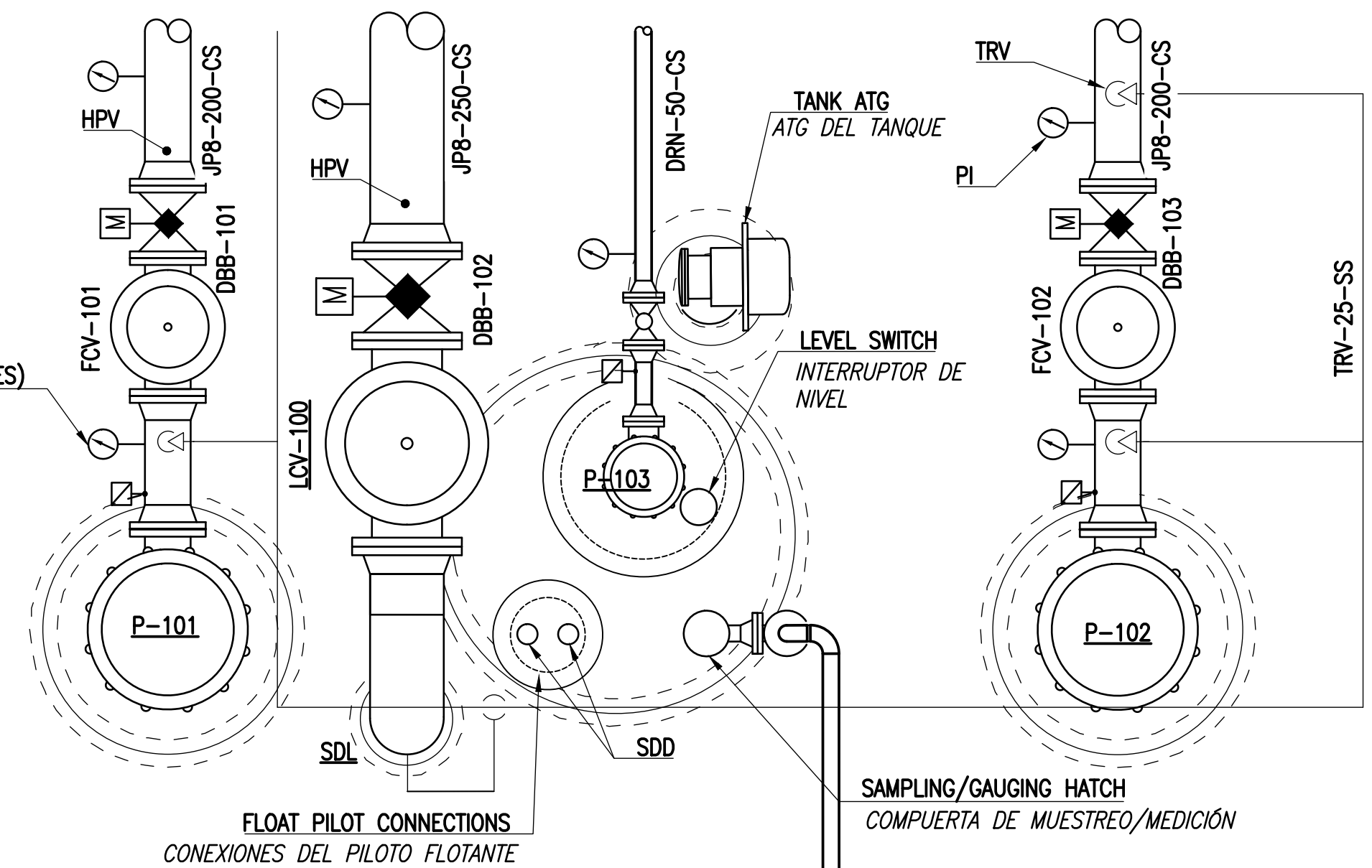
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SECTION
SECCIÓN
SCALE: 1:30
ESCALA: 1:30

A3

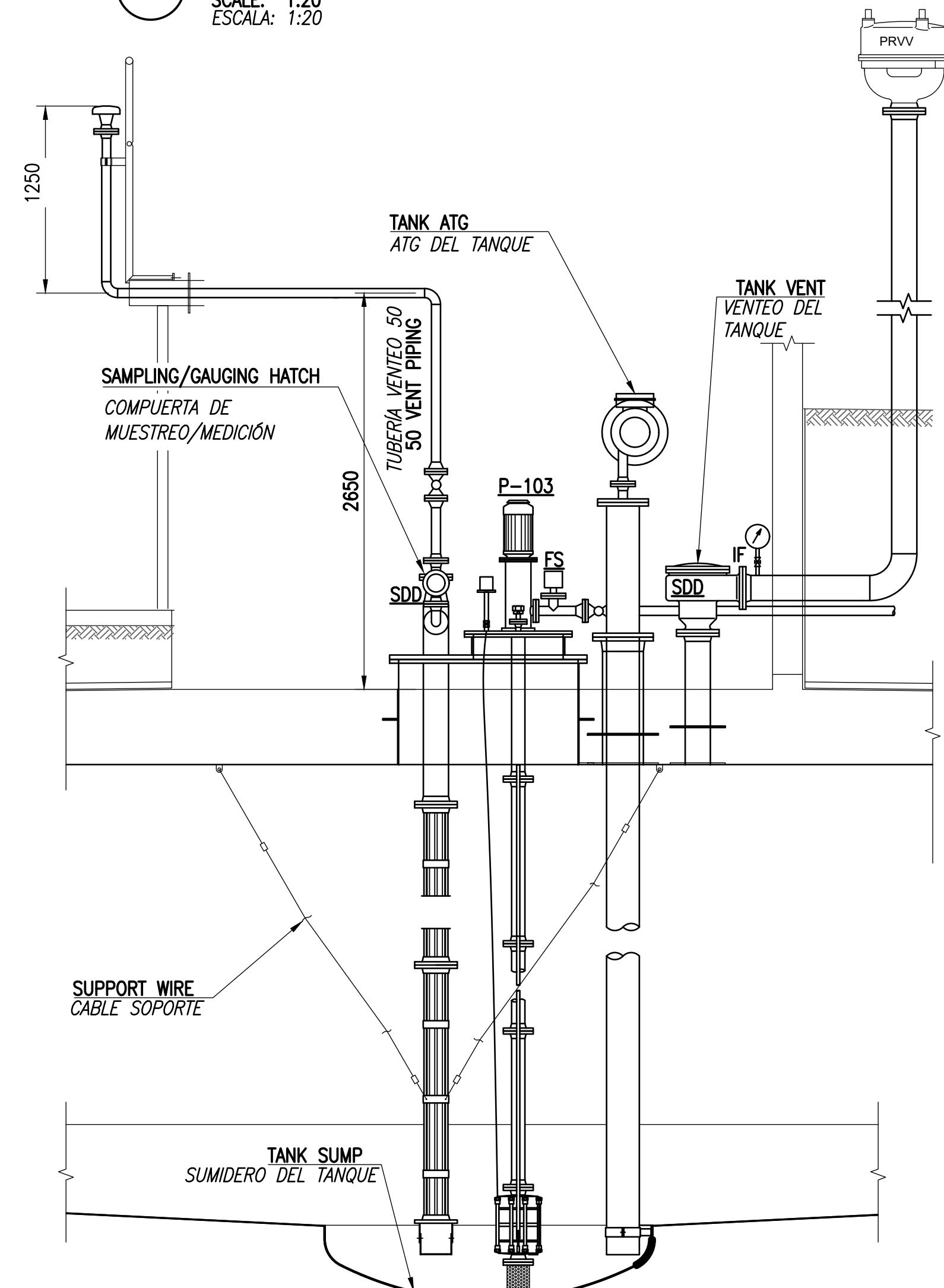
PRESSURE INDICATOR (TYP 6 PLACES)
INDICADOR DE PRESIÓN
(TIP EN 6 LUGARES)



TANK NOZZLES
TOBERAS DEL TANQUE

SCALE: 1:20
ESCALA: 1:20

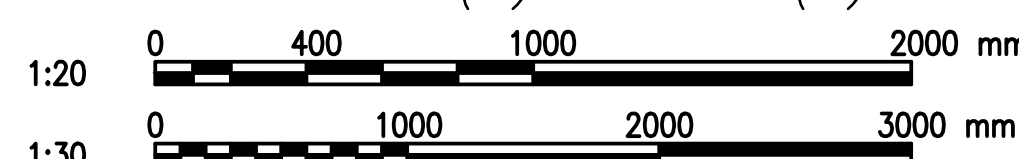
D4



SECTION
SECCIÓN
SCALE: 1:30
ESCALA: 1:30

A4

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):



DATE	APPR
SYN	DESCRIPTION
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO: DATE	
DES	BMS
DRW	RTH
CHK	DWN
PM/SM	CC/EA
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
ATLANTIC DESIGN AND CONSTRUCTION	
NAVAL STATION ROTA	
ROTA, SPAIN	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
PUMP VAULT PIPING PLAN	
PLANTA DE TUBERÍAS DE LA CÁMARA DEL TANQUE	
M-401T	
NAVFAC DRAWING NO. 14140182	
SHEET 78 OF 342	
NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018	

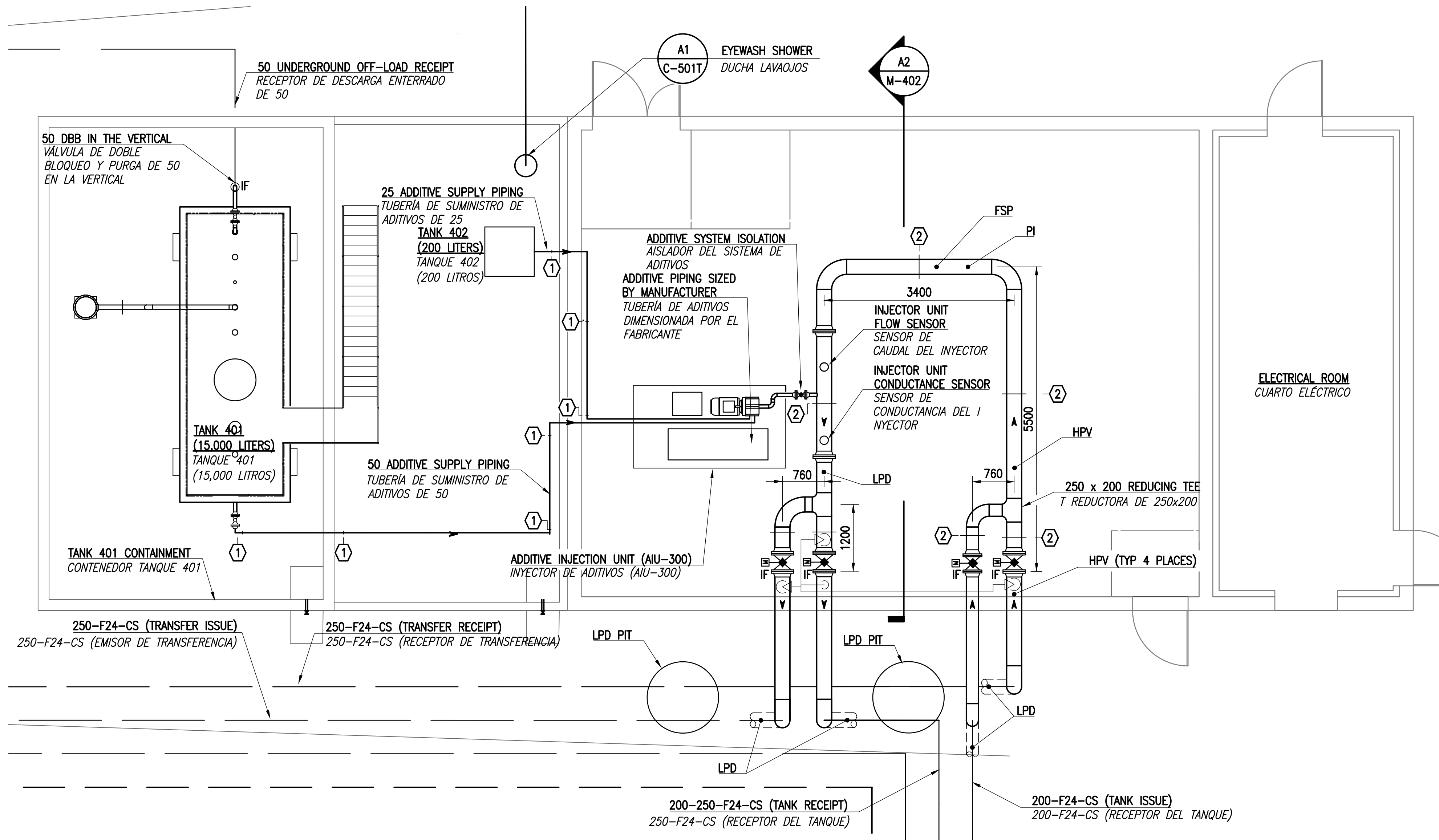
FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\NAM-402T ADDITIVE INJECTOR BUILDING.dwg LAYOUT NAME: m-402 PLOTTED: Tuesday, July 09, 2024 - 2:25pm USER: rhopkins

D

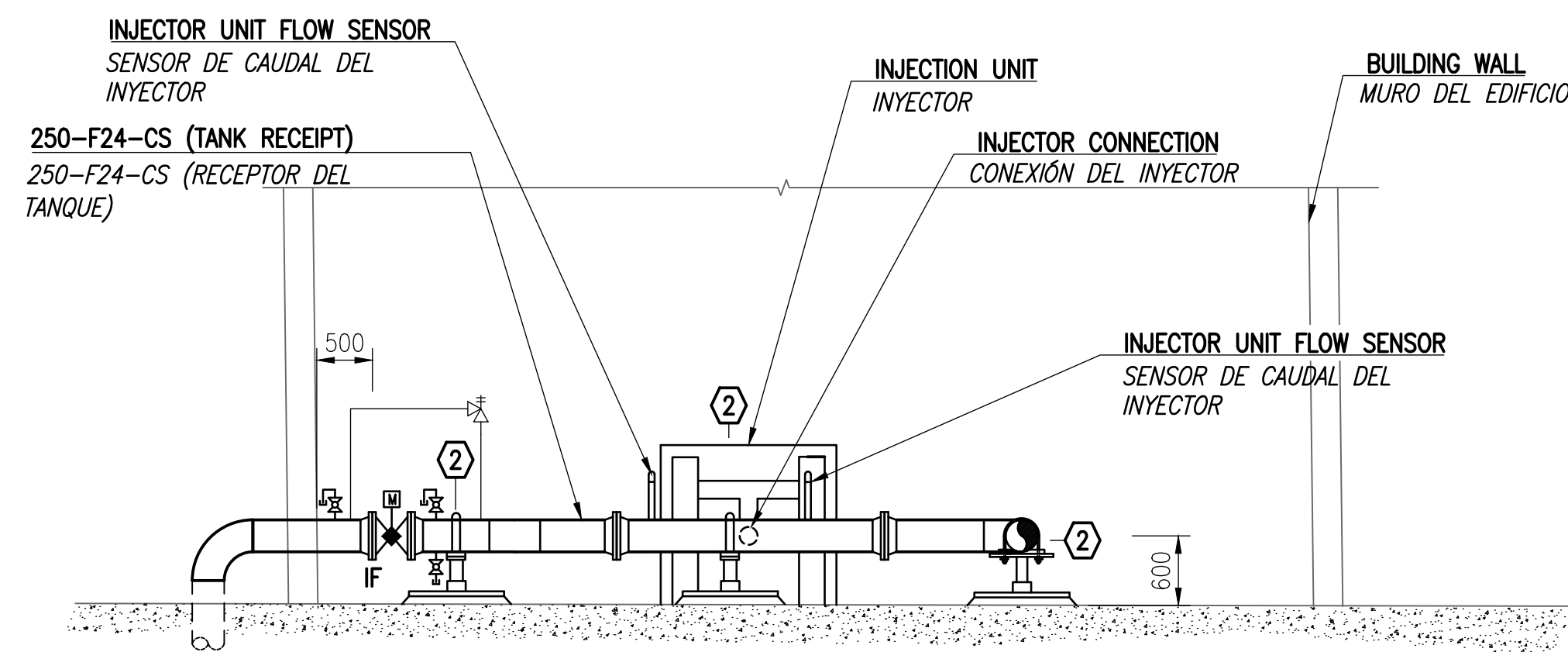
C

B

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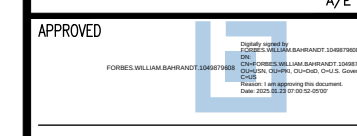
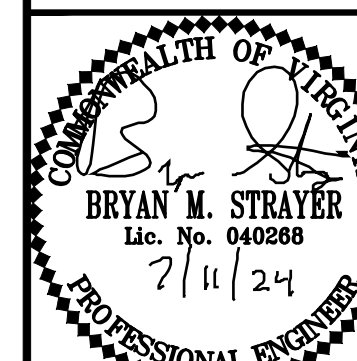
ADDITIVE INJECTOR BUILDING PIPING PLAN
PLANTA DE TUBERÍAS DEL EDIFICIO DEL INYECTOR DE ADITIVOS
SCALE: 1:50
ESCALA: 1:50



SECTION
SECCIÓN
A2
SCALE: 1:50
ESCALA: 1:50

ADDITIVE INJECTOR BUILDING NOTES: NOTAS DEL EDIFICIO DEL INYECTOR DE ADITIVOS:

1. THE ADDITIVE INJECTION EQUIPMENT, CONTROLS, AND ACCESSORIES, INCLUDING THE SMALL CI/LI ADDITIVE TANK, MUST BE PROVIDED AS A SINGLE SYSTEM UNIT FROM ONE SUPPLIER. THE FSII TANK WILL BE PROVIDED AS A STAND ALONE TANK, BUT WILL BE CONNECTED TO THE INJECTION SKID FOR PIPING AND CONTROLS. FSII TANK SUPPLIER WILL PROVIDE AN ADEQUATELY SIZED DESICCANT DRYER.
EL EQUIPO DE INYECCIÓN DE ADITIVOS, CONTROLES Y ACCESORIOS, INCLUYENDO LOS DOS TANQUES DE ADITIVOS PEQUEÑOS, DEBERÁN SER SUMINISTRADOS EN UN ÚNICO SISTEMA DESDE UN PROVEEDOR. EL TANQUE FSII SERÁ SUMINISTRADO COMO UN DEPÓSITO INDEPENDIENTE PERO ESTARÁ CONECTADO A LA PLATAFORMA DE INYECCIÓN PARA LAS TUBERÍAS Y CONTROLES.
2. ADDITIVE INJECTION SYSTEM WILL INCLUDE INJECTION PUMP SYSTEM WITH ALL ASSOCIATED PIPING AND CONTROLS. THIS SYSTEM WILL HAVE THE CAPACITY TO INJECT (3) DIFFERENT ADDITIVES INTO THE FUEL STREAM. THIS SYSTEM WILL ADDITIZE THE F-24 WITH CI/LI AND FSII. THE FINAL ADDITIVE INJECTION SLOT WILL BE UNUSED BUT AVAILABLE FOR FUTURE USE IF THE NEED REQUIRES.
EL SISTEMA DE INYECCIÓN DE ADITIVOS INCLUIRÁ EL SISTEMA DE BOMBA DE INYECCIÓN, Y TODAS LAS TUBERÍAS Y CONTROLES ASOCIADOS Y TENDRÁ CAPACIDAD PARA INYECTAR (3) ADITIVOS DIFERENTES EN EL CAUDAL DE COMBUSTIBLE. EL ADITIVO CI REQUERIRÁ UN DEPÓSITO DE ALMACENAMIENTO DEDICADO QUE SE PROVEERÁ COMPLETO COMO PARTE DEL SISTEMA DE INYECCIÓN DE ADITIVOS. EL ADITIVO FSII SERÁ ALMACENADO EN UN TANQUE AÉREO CERCANO Y SEPARADO.



FOR COMMANDER NAVFAC
ACTIVITY

SATISFACTORY TO: DATE

DES: BMS | DRW: RTH | CHK: DWN
PM/CM: CCA/CLA

BRANCH MANAGER
DESIGN MANAGER

FIRE PROTECTION

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC

NAVAL STATION ROTA
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

ADDITIVE INJECTOR BUILDING PIPING PLAN
PLANTA DE TUBERÍAS DEL EDIFICIO DEL INYECTOR DE ADITIVOS

SCALE:

PROJECT NO.:

CONSTR. CONTR. NO.:

NAVFAC DRAWING NO.:

14140183

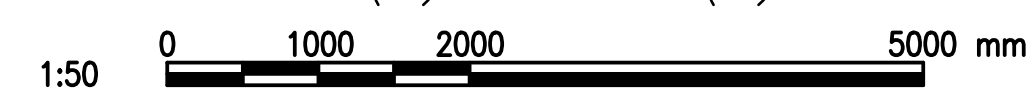
SHEET 79 OF 342

M-402T

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

GRAPHIC SCALE(S):

ESCALA(S) GRÁFICA(S):



FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\M-M-405T MECHANICAL TANK PLAN.dwg LAYOUT NAME: M-405T PLOTTED: Tuesday, July 09, 2024 - 2:27pm USER: rhopkins

D

C

B

A

15,000-LITERS TANK NOZZLE SCHEDULE

CUADRO BOQUILLA TANQUE 15.000 LITROS

- (A) #50 FILL LINE (WITH OVERFILL PREVENTION VALVE)
LINEA DE LLENADO #50 (CON VALVULA DE PREVENCIÓN DE REBOSE)
- (B) #100 ATG (WITH INTEGRAL LEVEL ALARMS)
ATG #100 (CON ALARMAS DE NIVEL INTEGRADAS)
- (C) #50 CLOCK-TYPE MECHANICAL GAUGE
MANÓMETRO MECÁNICO TIPO RELOJ #50
- (D) #100 VENT TO DESICCANT DRYER
VENTEO #100 HASTA SECADOR DESECANTE
- (E) #100 MANUAL GAUGE/SAMPLING HATCH
MANÓMETRO MANUAL/TRAMPILLA MUESTREO #100
- (F) 762mm MANWAY
BOCA DE ACCESO PARA PERSONAL 762mm
- (G) EMERGENCY VENT OPENING
HUECO DE VENTEO DE EMERGENCIA
- (H) #50 OUTLET
SALIDA #50
- (I) #50 WATER DRAW-OFF
EVACUACIÓN DE AGUA #50
- (J) LEVEL SWITCHES
INTERRUPTORES DE NIVEL

TANK 401 NOTES:

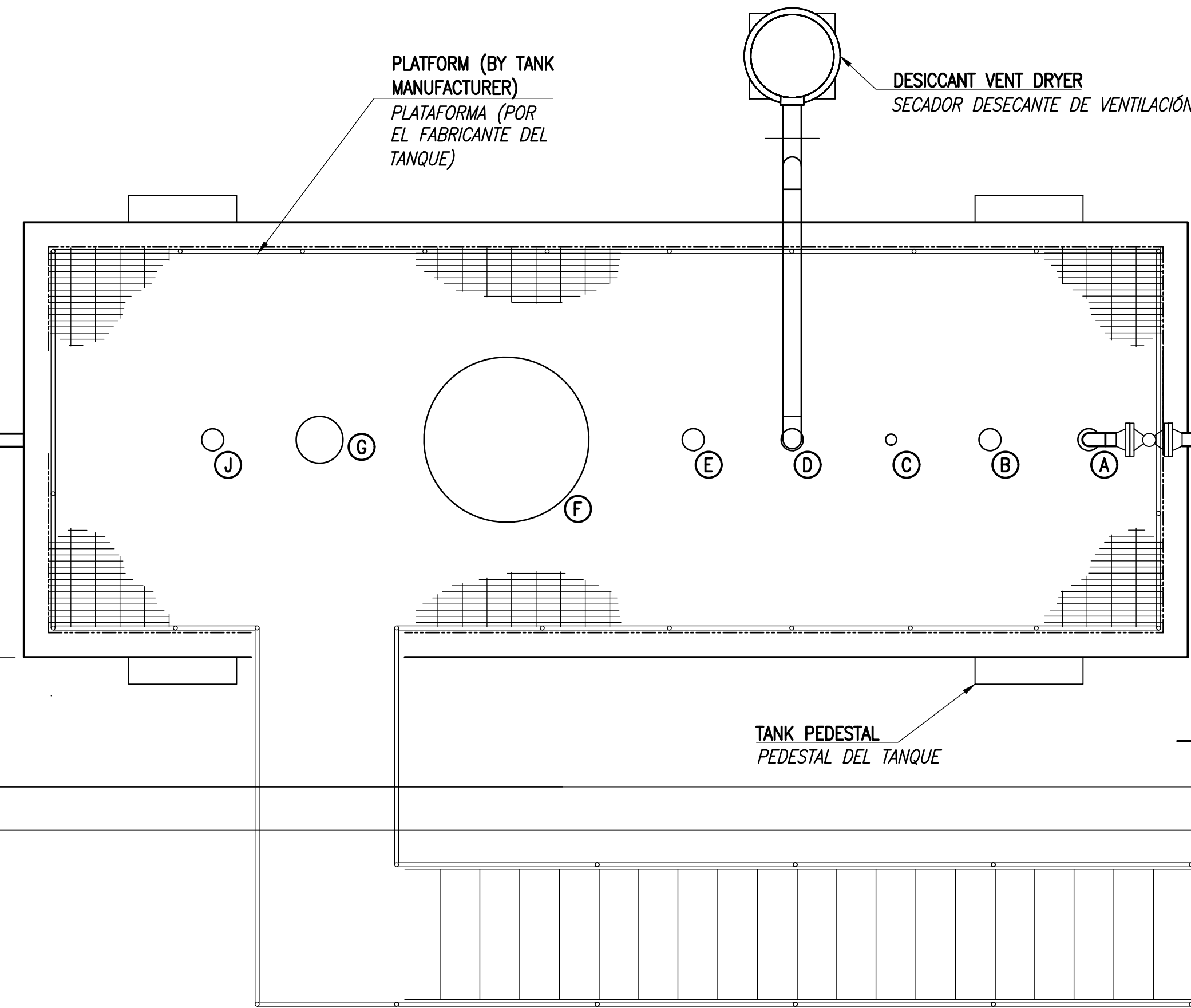
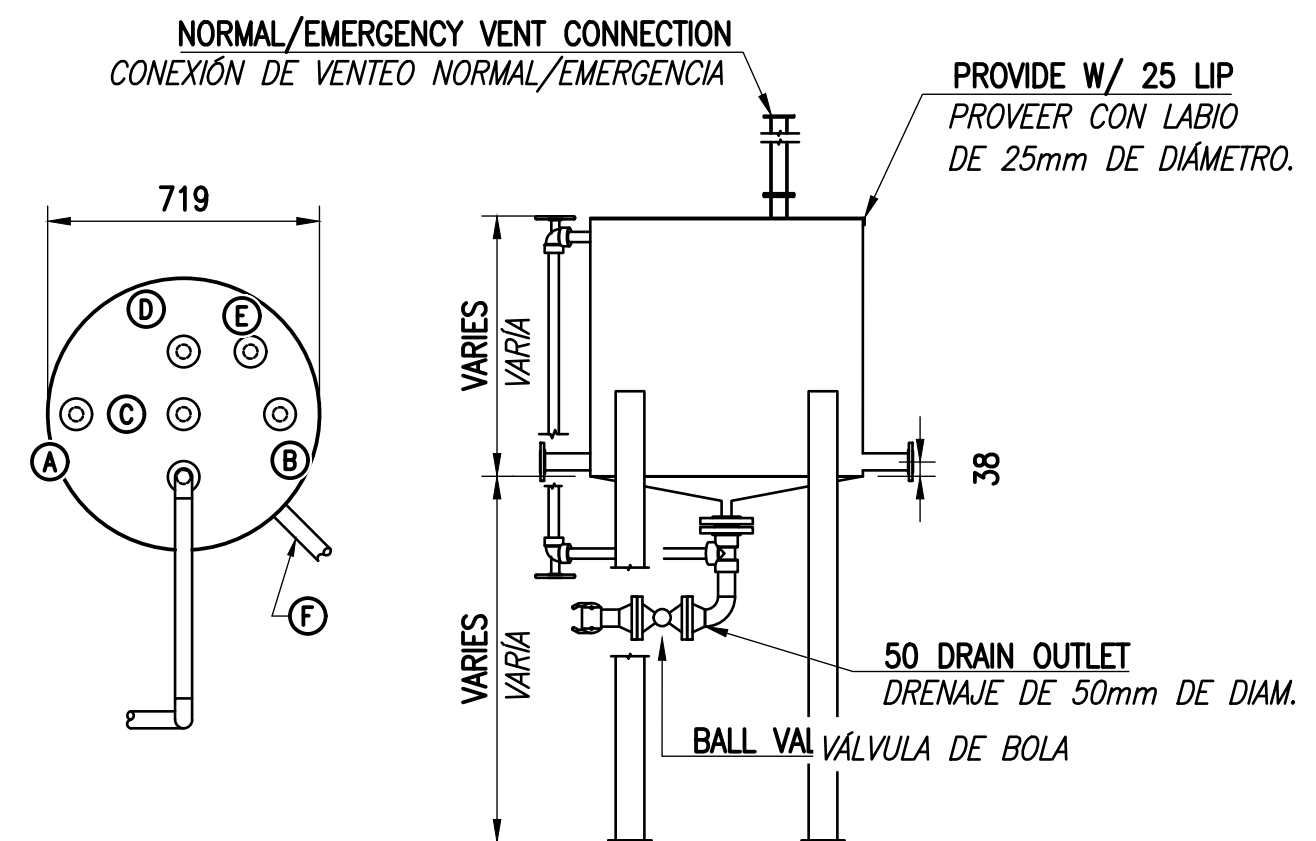
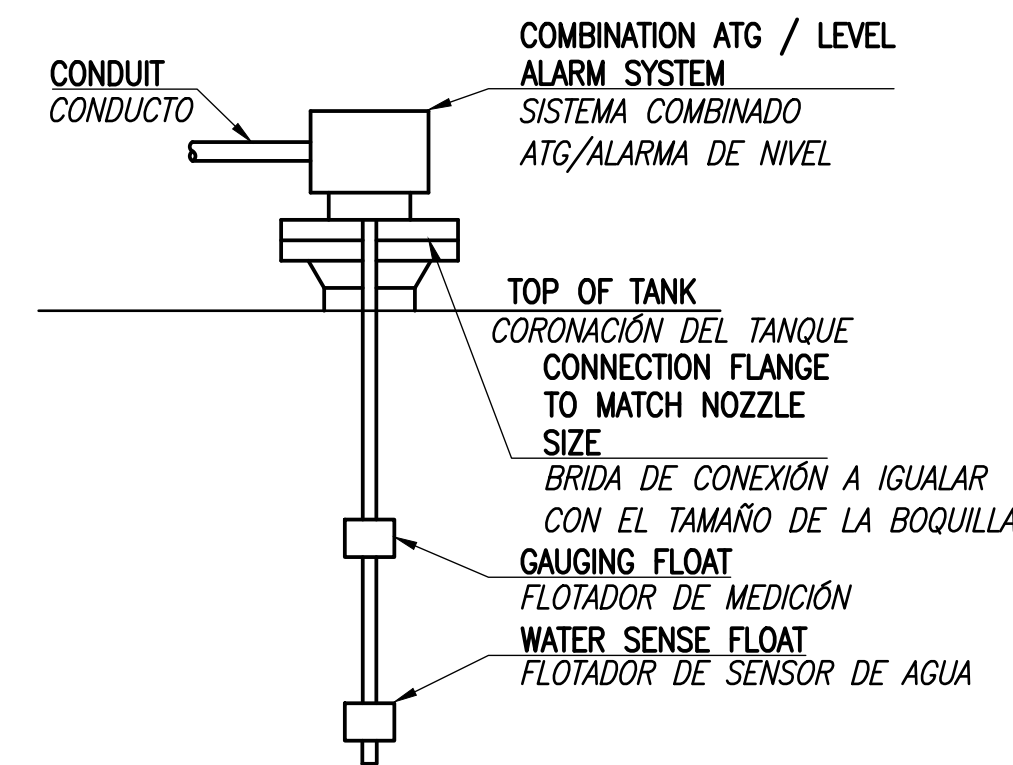
NOTAS TANQUE 401:

- LEVEL OF PLATFORM ABOVE TANK MUST BE AT LEVEL OF TANK MANHOLE FLANGES.
1. EL NIVEL DE LA PLATAFORMA SOBRE EL TANQUE DEBERÁ ESTAR A NIVEL CON LAS BRIDAS DEL POZO DEL TANQUE.
- PROJECTION OF FILL NOZZLE AND VENT LINE TO BE 150mm MINIMUM ABOVE PLATFORM.
2. LA PROYECCIÓN DE LA BOQUILLA DE LLENADO Y LA LINEA DE VENTEO ESTARÁ A 150mm MÍNIMO SOBRE LA PLATAFORMA.
- PLATFORM MUST PROVIDE ACCESS TO ALL APPURTENANCES ON TOP OF THE TANKS.
3. LA PLATAFORMA DEBERÁ PROVEER ACCESO A TODOS LOS ELEMENTOS EN LA CORONACIÓN DE LOS TANQUES.
- TANK ACCESS PLATFORM, INTERCONNECTING PLATFORMS, RAILINGS, AND STAIRS INDICATED MUST BE DESIGNED, COORDINATED AND PROVIDED BY THE TANK MANUFACTURER.
4. LA PLATAFORMA DE ACCESO DEL TANQUE, PLATAFORMAS DE INTERCONEXIÓN, BARANDILLAS Y ESCALERAS INDICADAS DEBERÁN ESTAR DISEÑADAS, COORDINADAS Y SUMINISTRADAS POR EL FABRICANTE DEL TANQUE.
- TANK PLATFORM/ STAIRS/ HANDRAILS/ GUARDRAILS MUST BE DESIGNED IN ACCORDANCE WITH OSHA STANDARDS AND 100 PSF LIVE LOAD.
5. LA PLATAFORMA DE ACCESO DEL TANQUE, PLATAFORMAS DE INTERCONEXIÓN, BARANDILLAS Y ESCALERAS INDICADAS DEBERÁN ESTAR DISEÑADAS, COORDINADAS Y SUMINISTRADAS POR EL FABRICANTE DEL TANQUE.

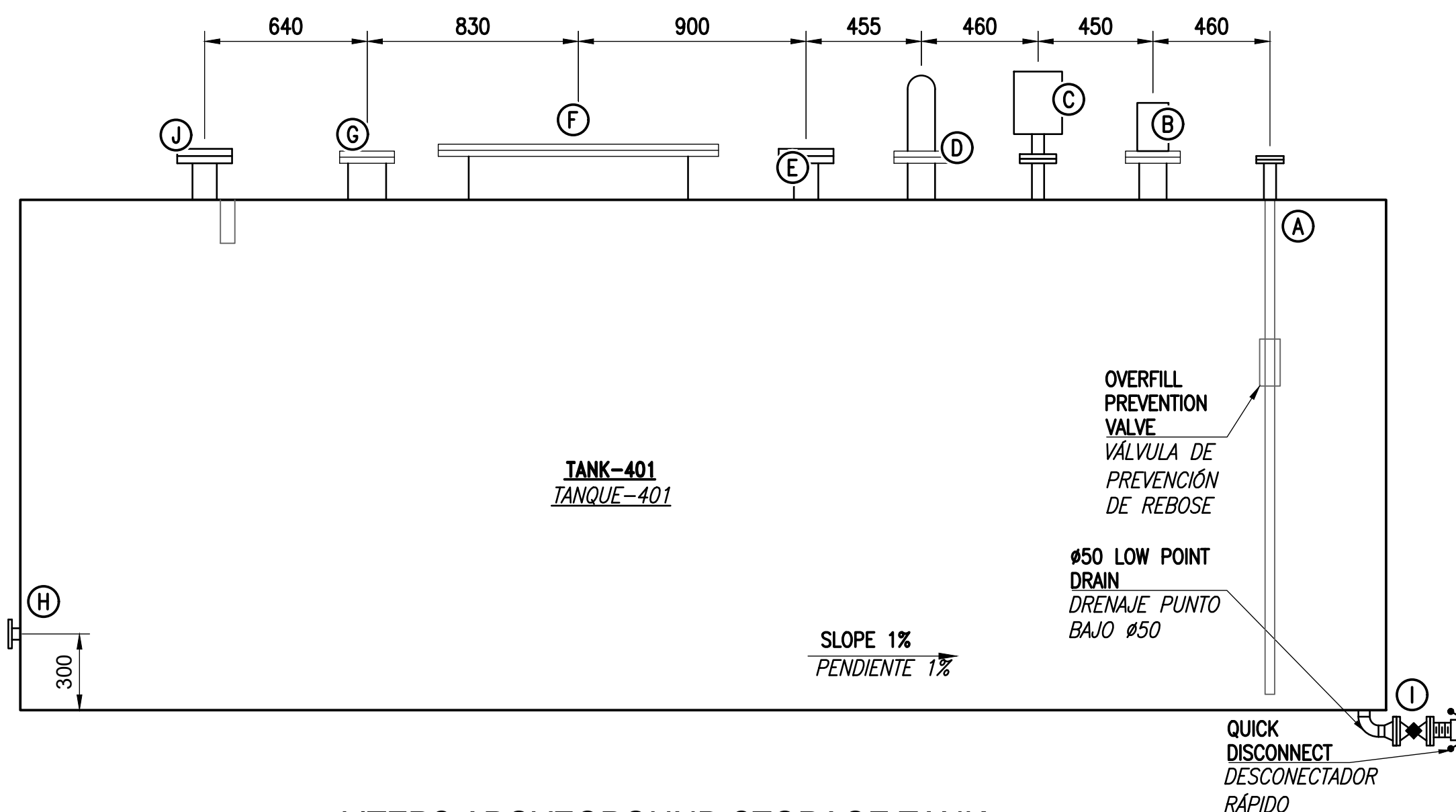
200-LITER TANK NOZZLE SCHEDULE

CUADRO BOQUILLA TANQUE 200 LITROS

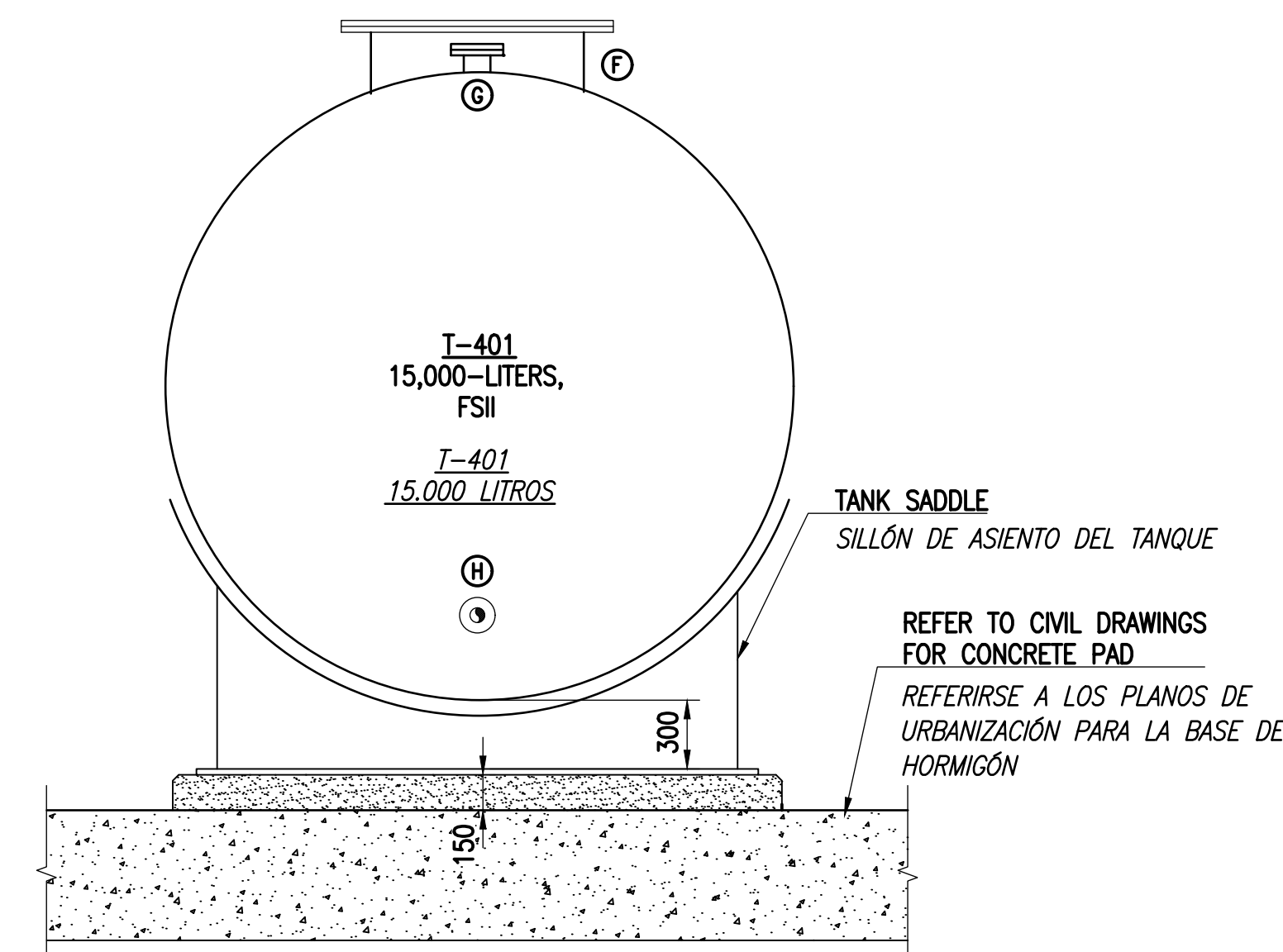
- (A) #50 FILL LINE
LINEA DE LLENADO #50
- (B) MANUAL ADDITIVE FILL
RELLENO DE ADITIVO MANUAL
- (C) #50 MIXER
MEZCLADOR #50
- (D) #50 OPEN ATMOSPHERIC VENT
VENTO ABIERTO A LA ATMÓSFERA #50
- (E) #50 OUTLET
SALIDA #50
- (F) #50 LOW POINT DRAIN
DRENAJE DE PUNTO BAJO #50



15,000 LITERS ABOVEGROUND STORAGE TANK PLAN
PLANTA DE TANQUE DE ALMACENAJE SOBRE EL TERRENO 15.000 LITROS



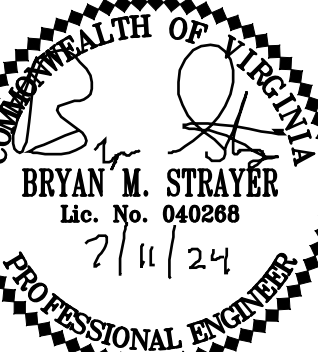
15,000 LITERS ABOVEGROUND STORAGE TANK
TANQUE DE ALMACENAJE ENTERRADO DE 15.000 LITROS



GRAPHIC SCALE(S):

ESCALA(S) GRÁFICA(S):

0 400 1000 2000 mm



FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DESIGNED BY: BMS DRAWN BY: RTH CHECKED BY: DWN

IN/OUT: CCA/CLA

DESIGN MANAGER

FIRE PROTECTION

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC

ATLANTIC DESIGN AND CONSTRUCTION

NAVAL STATION ROTA

ROTA, SPAIN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

MECHANICAL TANK PLAN

PLANTA MECÁNICA DEL TANQUE

SCALE:

PROJECT NO.:

CONSTR. CONTR. NO.:

NAVFAC DRAWING NO.:

14140186

SHEET

OF 342

M-405T

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\M-M-501T DETAILS.dwg LAYOUT NAME: M-501T DETAILS PLOTTED: Tuesday, July 09, 2024 - 1:42pm USER: hopkins

D

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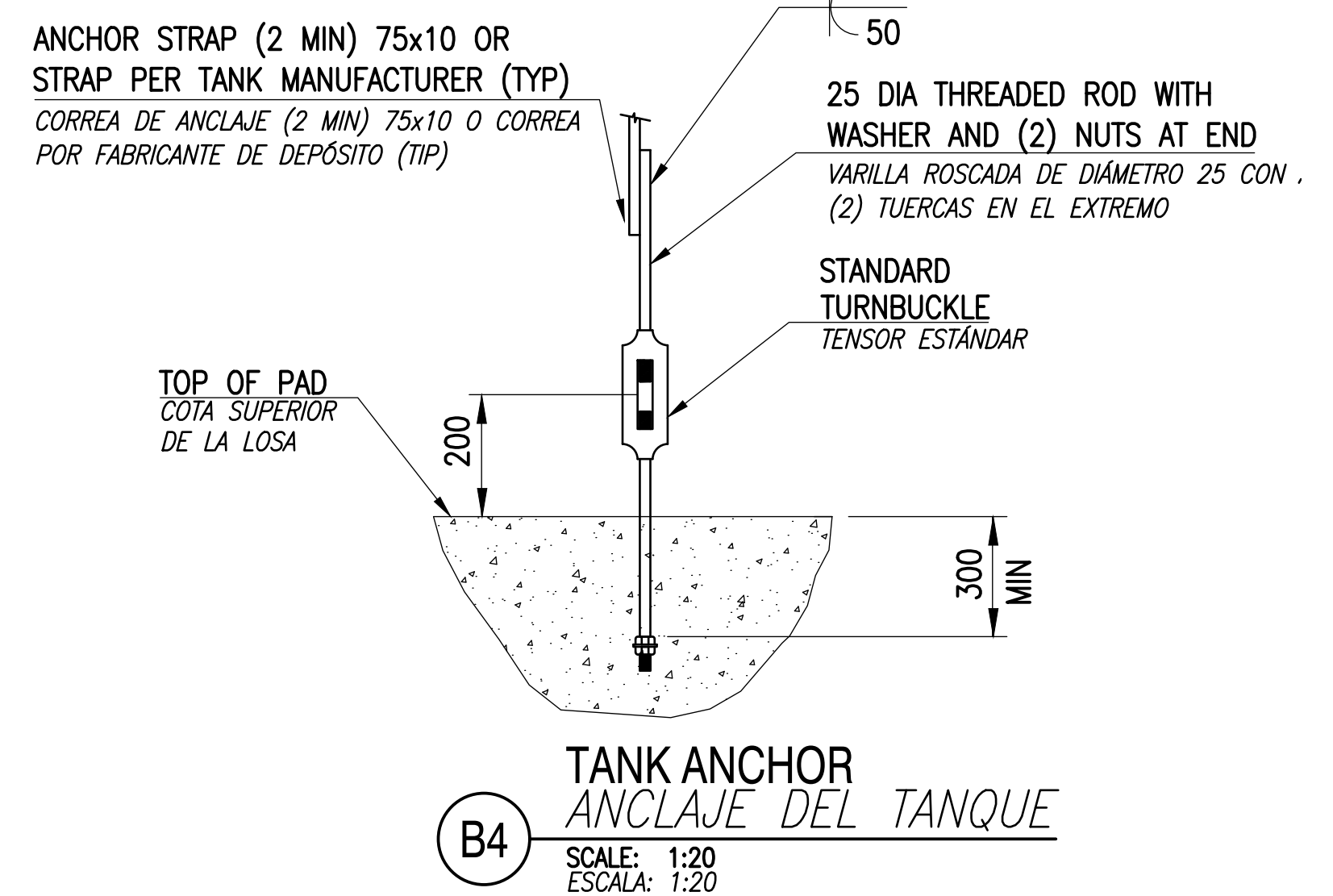
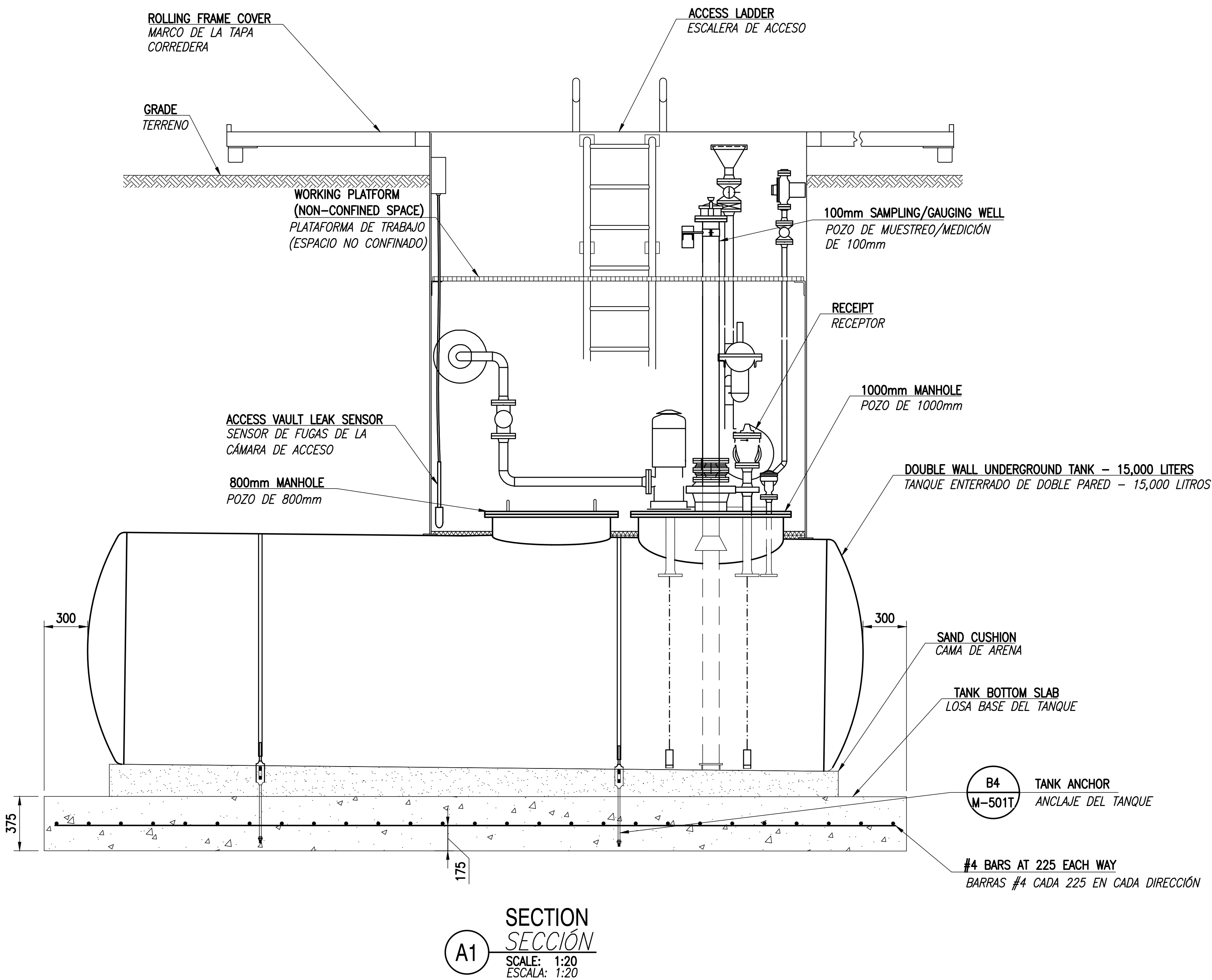
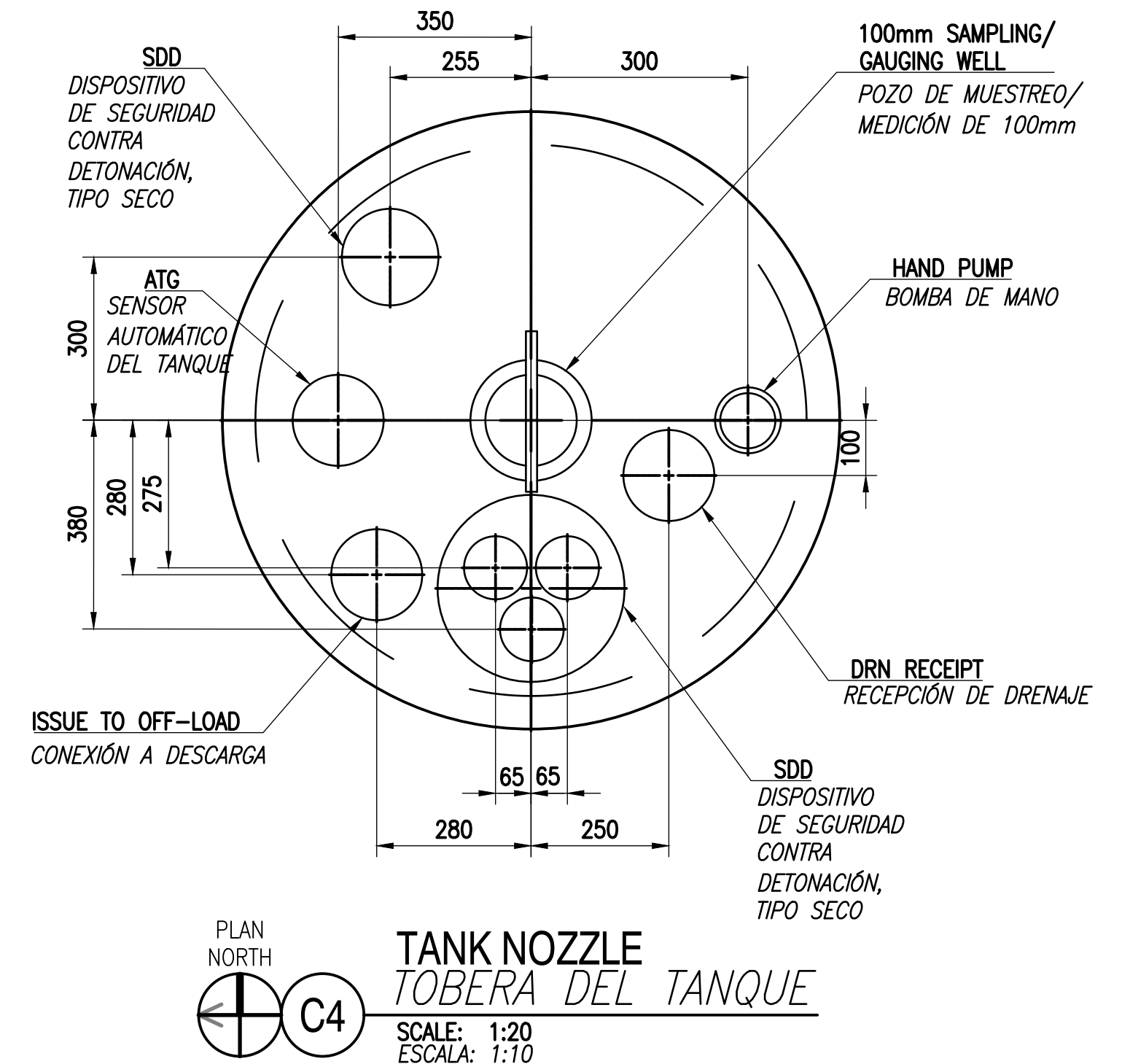
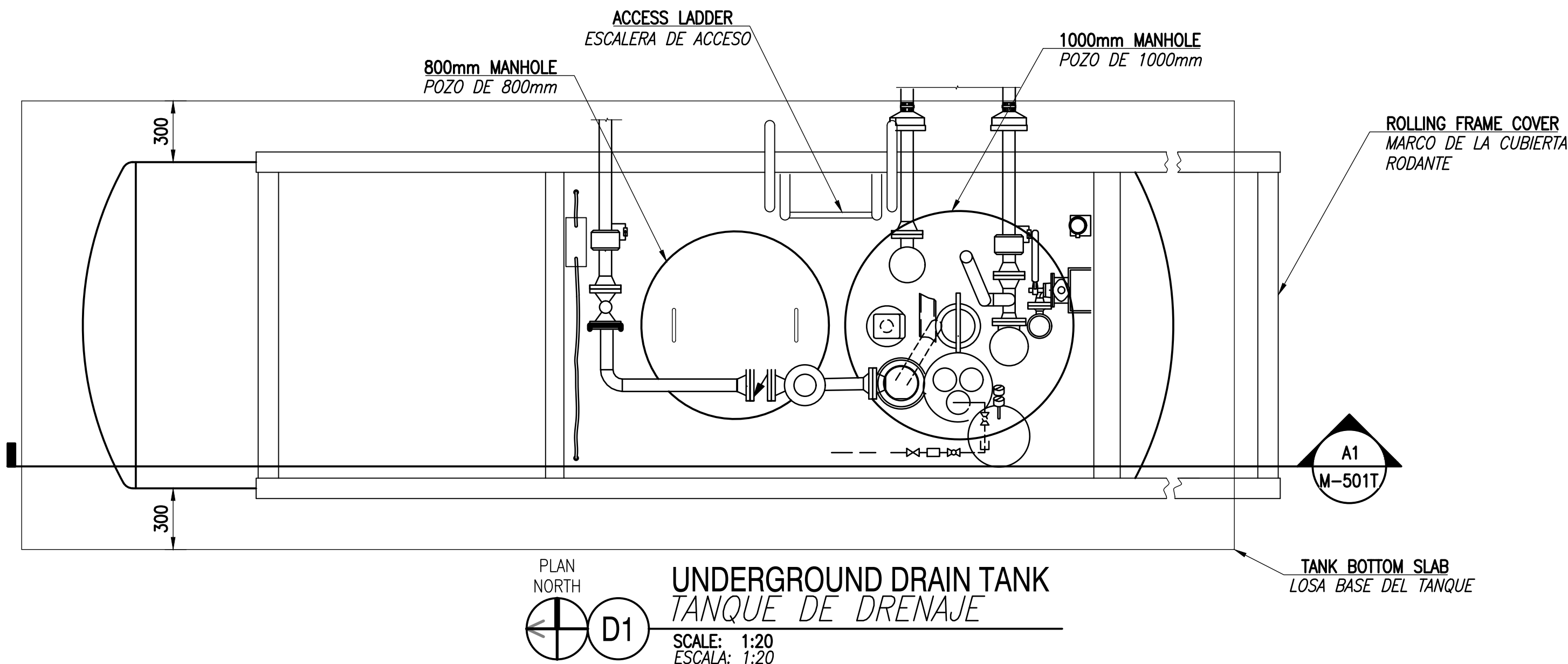
A

D

C

B

A

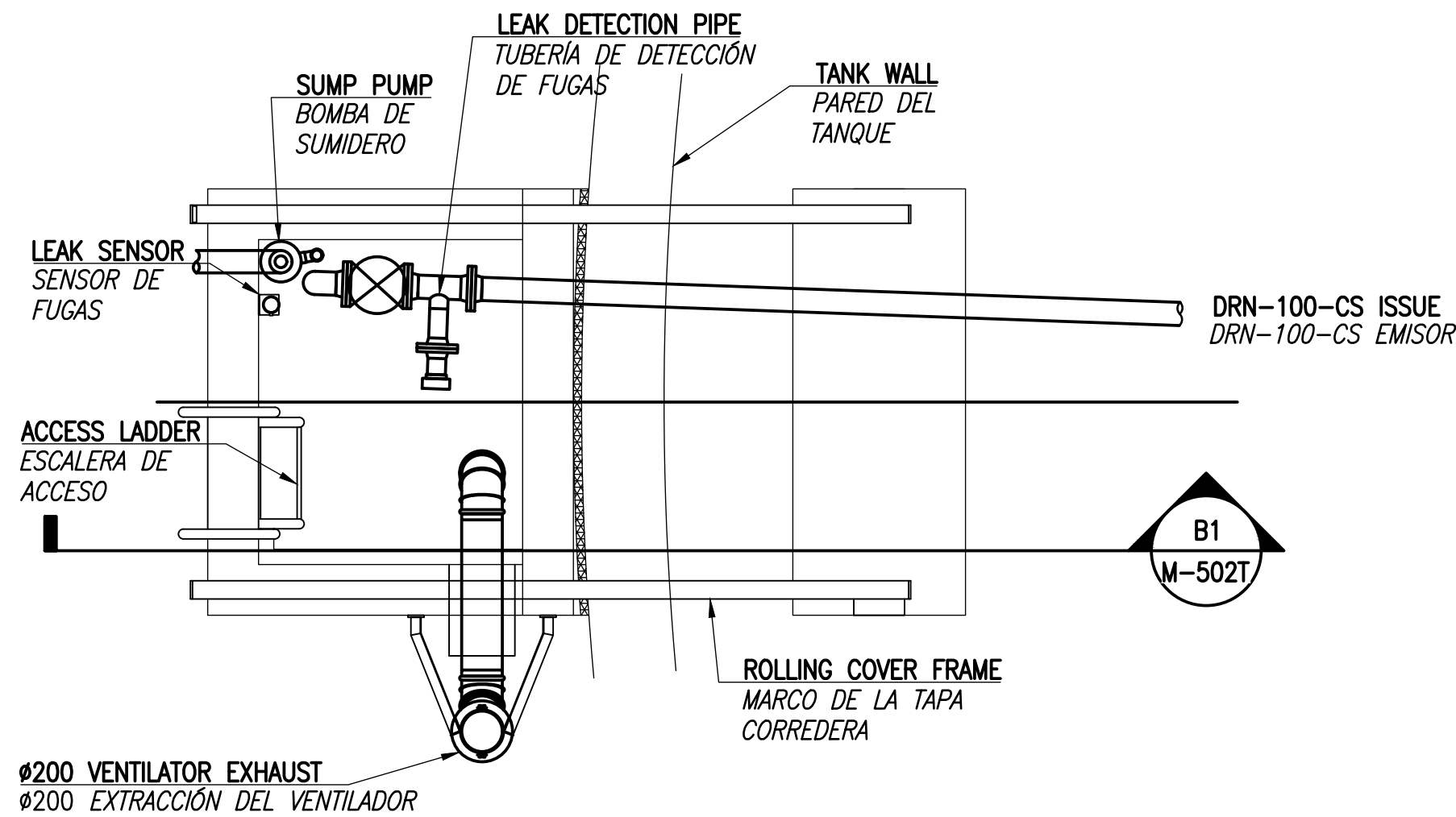


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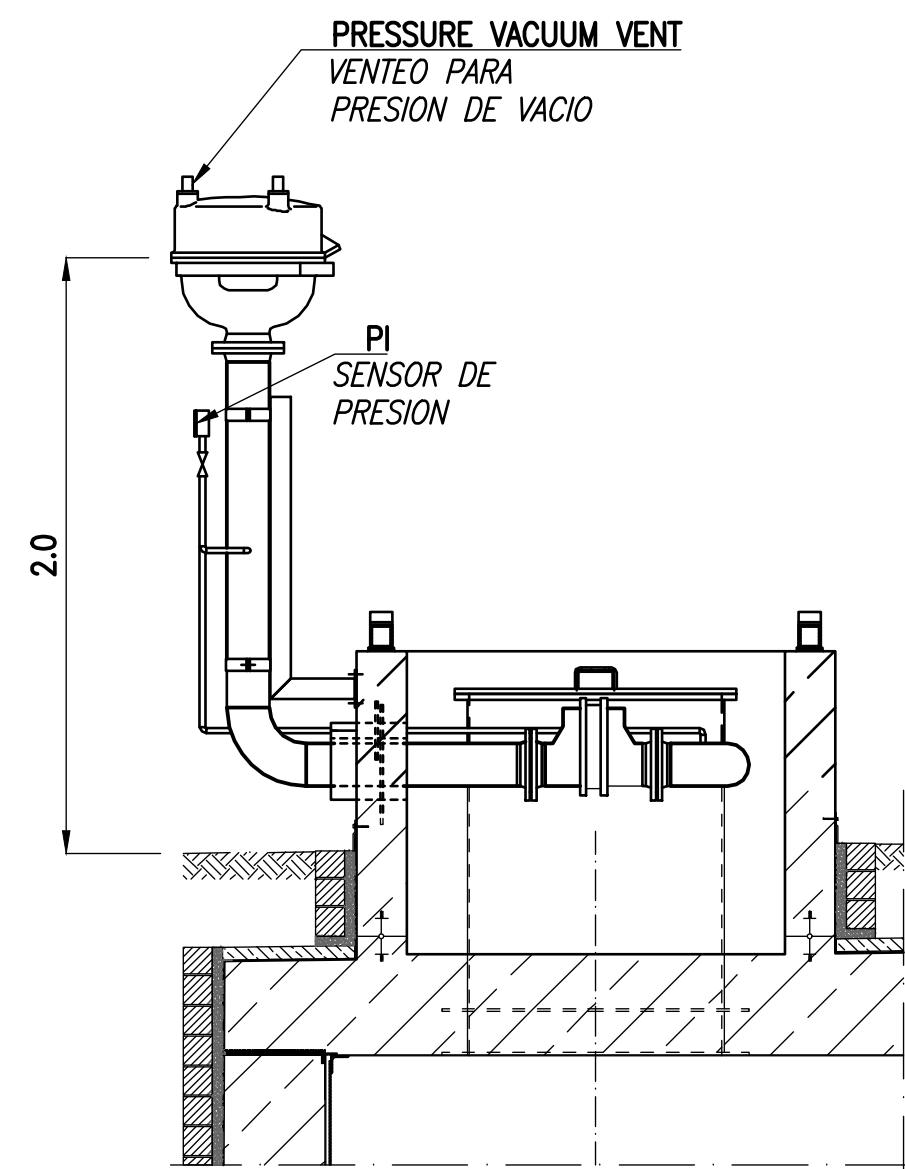
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DATE	APPR
SYN	DESCRIPTION
APPROVED	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO: DATE	
DES	BMS
DRAW	RTH
CHK	DWN
PM/SM	CC/EA
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
ATLANTIC DESIGN AND CONSTRUCTION	
NAVAL STATION ROTA	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
ROTA, SPAIN	
DETAILS	
SCALE:	
PROJECT NO.:	
CONSTR. CONTR. NO.:	
NAVFAC DRAWING NO.:	
14140187	
SHEET	
83 OF 342	
M-501T	
NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018	

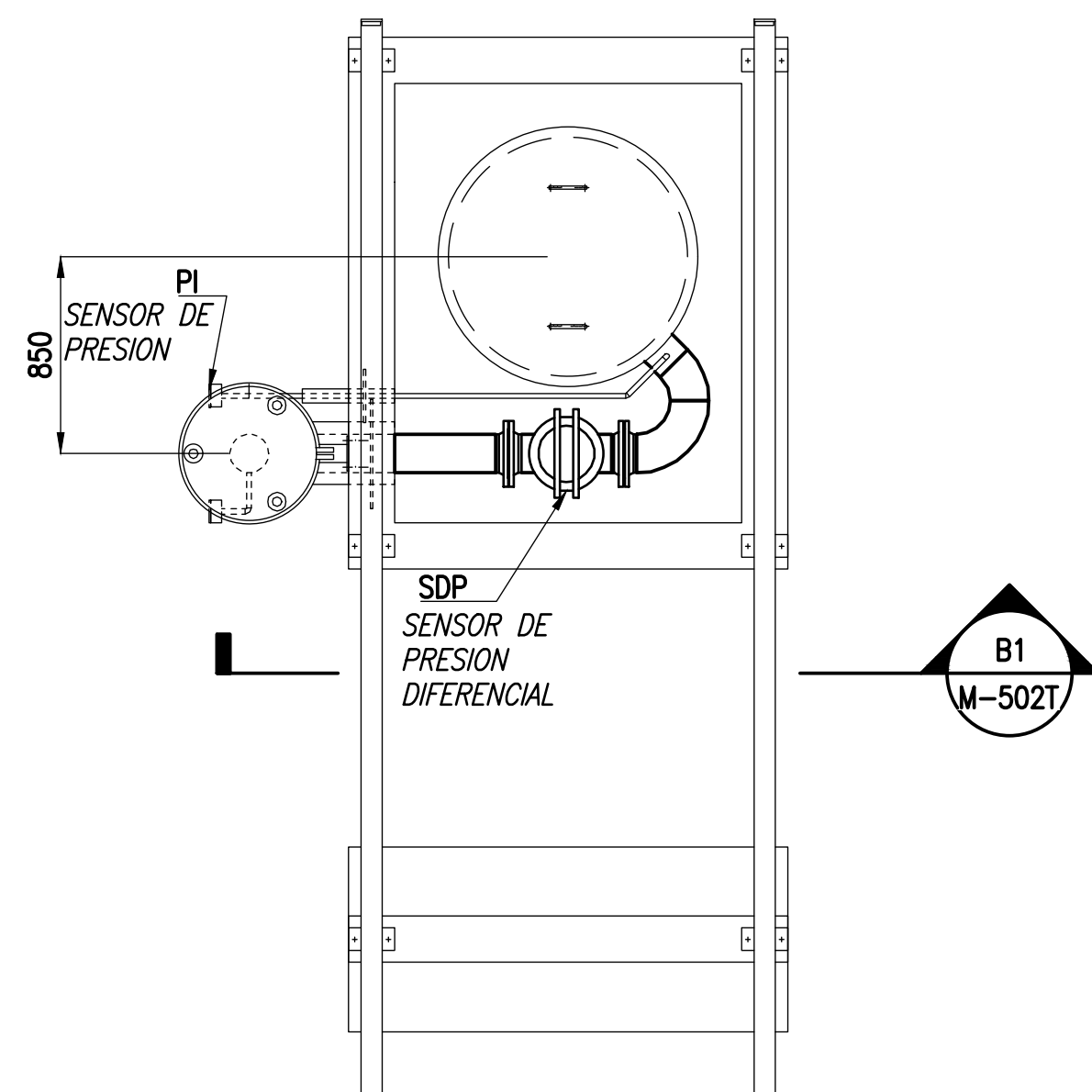
FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tank - Rota\CAD\M-M-502T DETAILS.dwg LAYOUT NAME: M-502T DETAILS PLOTTED: Tuesday, July 09, 2024 - 1:42pm USER: Hopkins



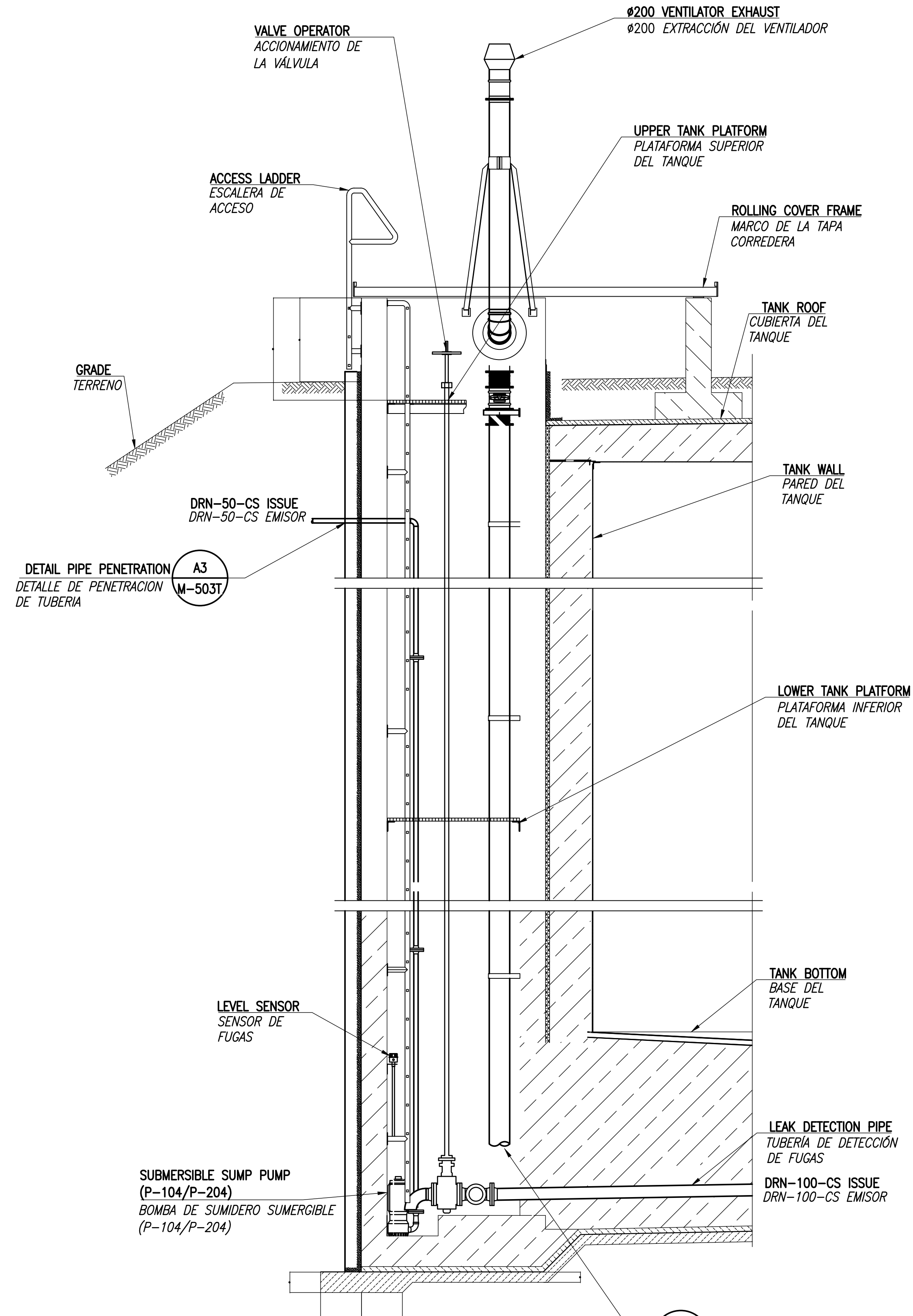
D1 LEAK DETECTION VAULT
CÁMARA DE DETECCIÓN DE FUGAS
SCALE: 1:30
ESCALA: 1:30



B1 SECTION
SECCIÓN
SCALE: NONE
ESCALA: TT



B1 MANHOLE WITH TANK VENT PIPING
ARQUETA CON TUBERÍA DE VENTEO DEL TANQUE
SCALE: 1:30
ESCALA: 1:30

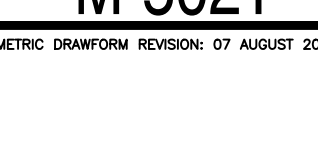
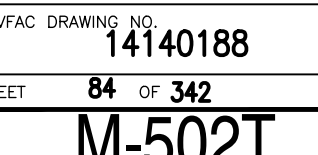
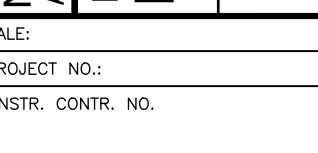
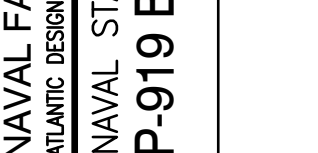
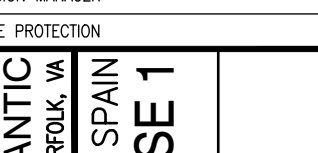
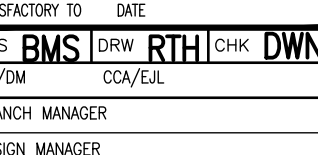
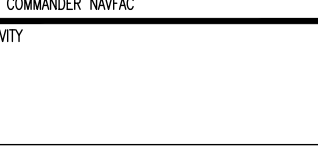
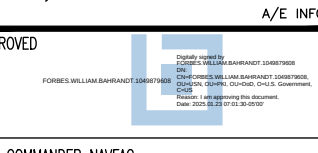
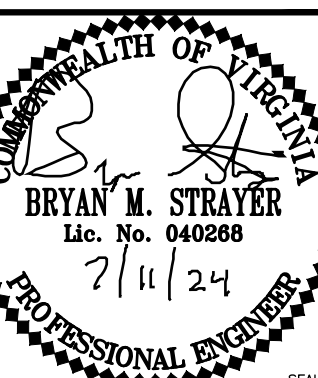


A1 SECTION
SECCIÓN
SCALE: 1:30
ESCALA: 1:30

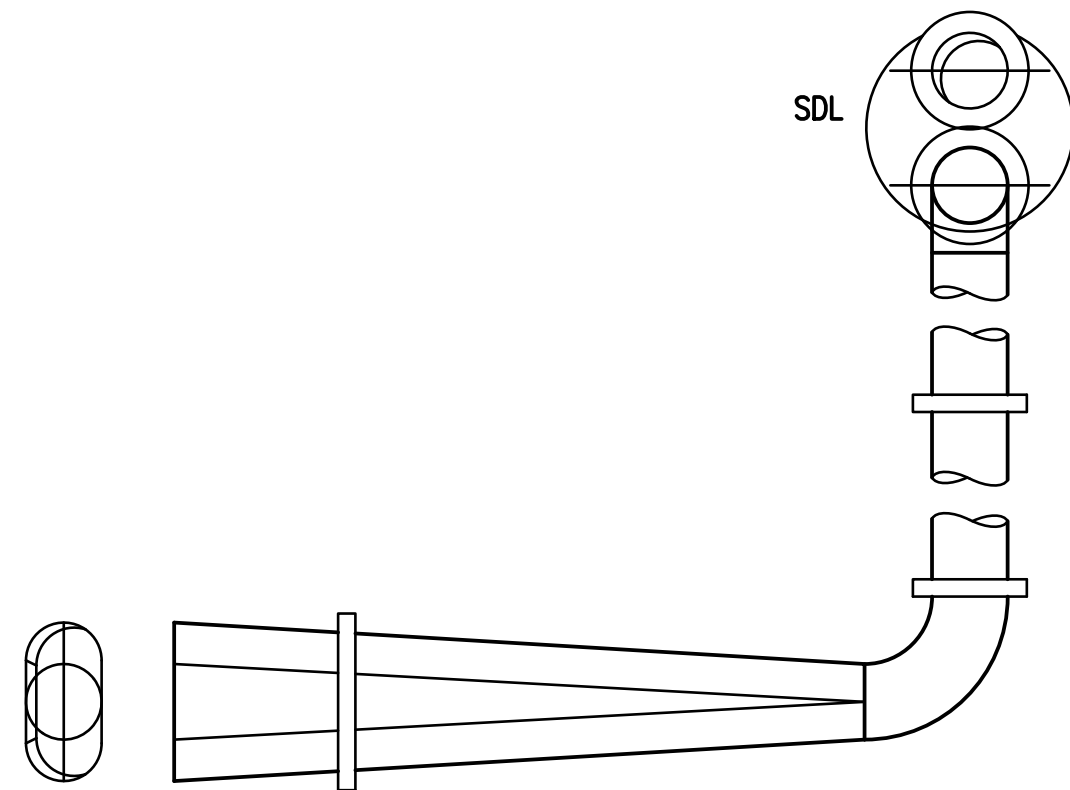
NOTE: REFER TO M-201T FOR DRAIN PIPING DETAILS.
NOTAS: REFERIRSE A PLANO M-201T PARA DETALLES DE TUBERÍA DE DRENAJE

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):

1:30 0 1000 2000 3000 mm



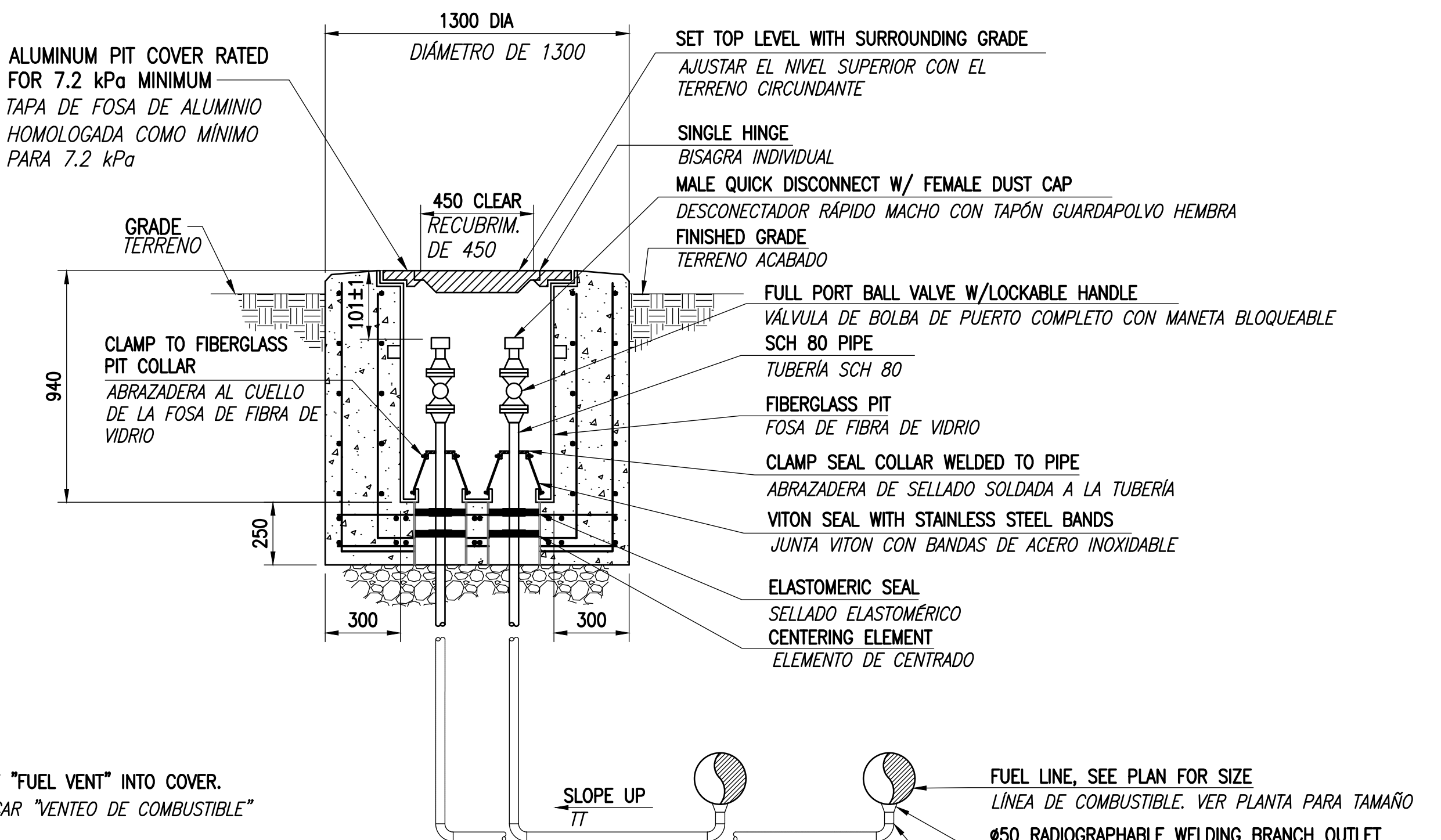
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RECEIPT NOZZLE

BOQUILLA DE RECEPCIÓN

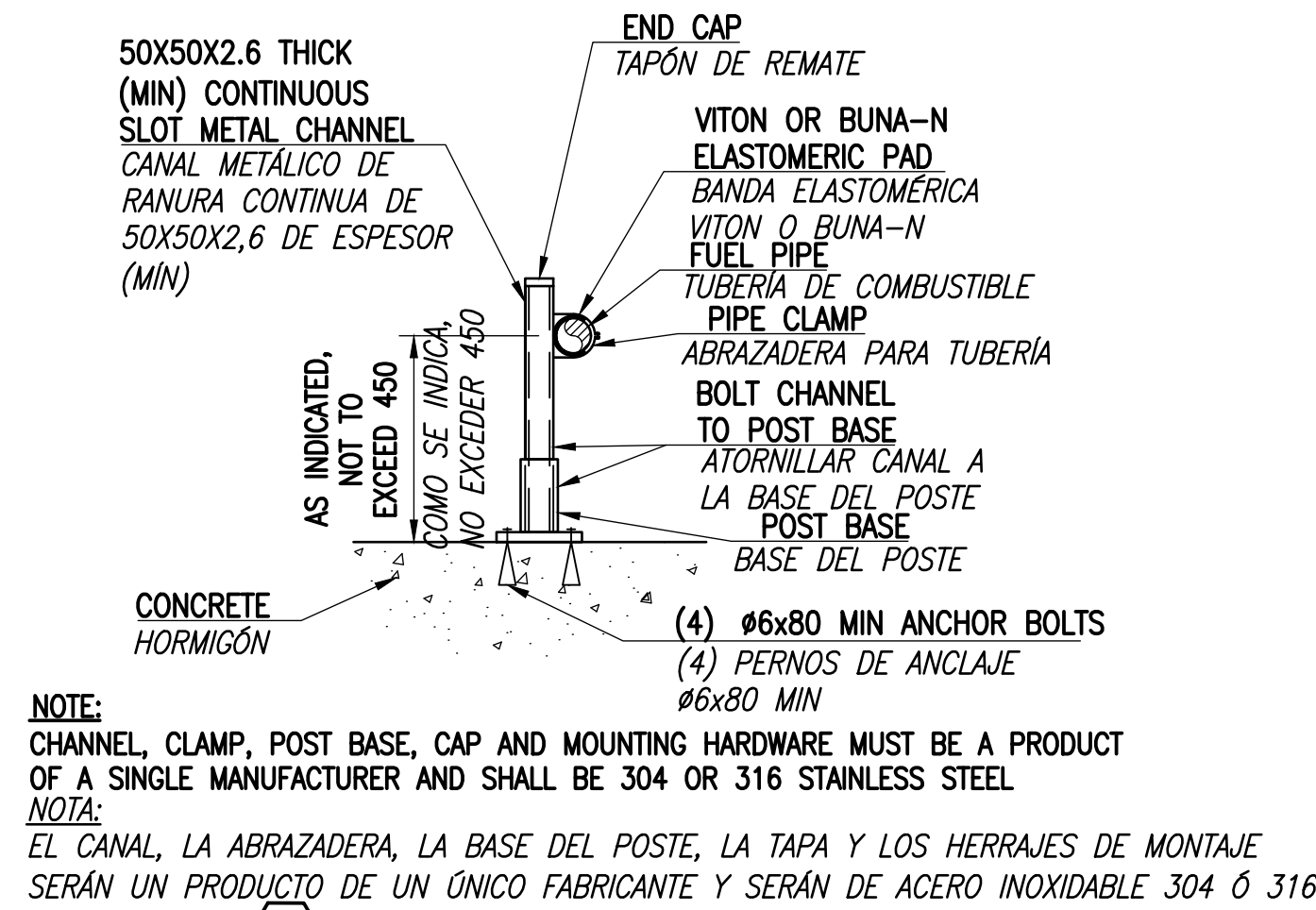
SCALE: NONE
ESCALA: SIN ESCALA



LOW POINT DRAIN (LPD) PIT

FOSA DE DRENAJE EN PUNTO BAJO (LPD)

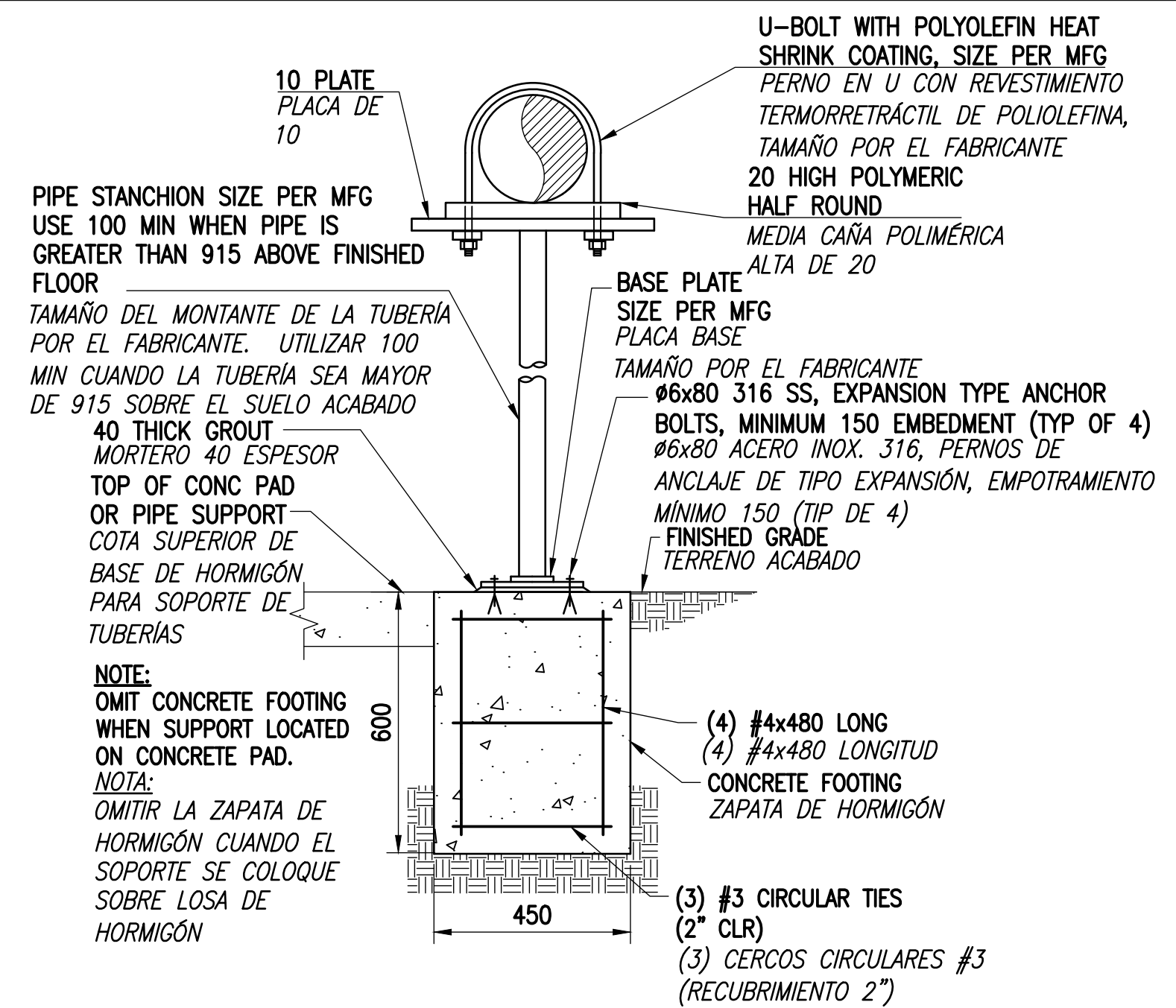
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ESCALA: SIN ESCALA



TYPE 1 SMALL PIPE SUPPORT

SOPORTE TUBERÍA PEQUEÑA TIPO 1

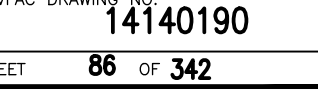
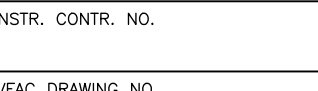
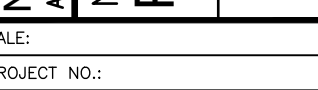
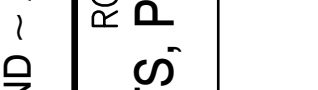
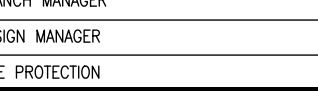
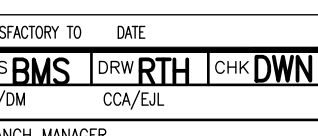
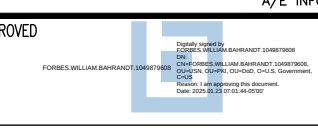
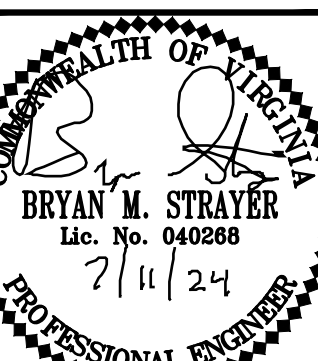
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ESCALA: SIN ESCALA



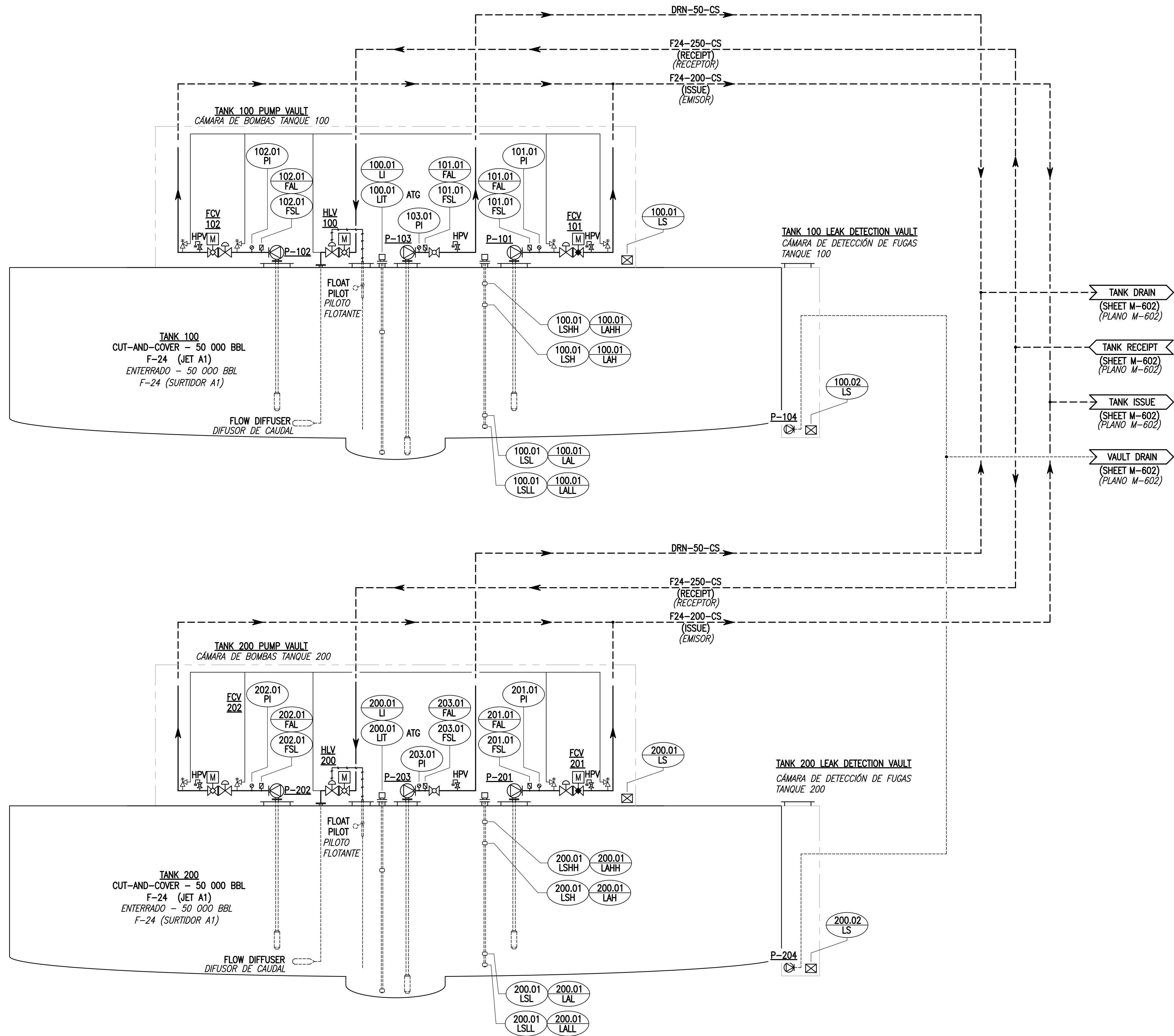
TYPE 2 FLOOR SUPPORTED LARGE PIPE SUPPORT

SOPORTE TUBERÍA LARGA APOYADO EN SUELO TIPO 2

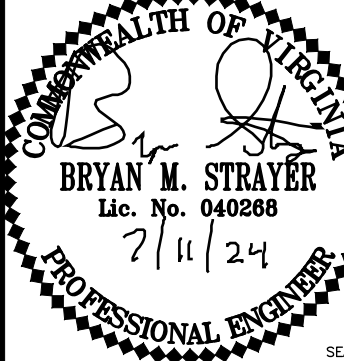
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ESCALA: SIN ESCALA



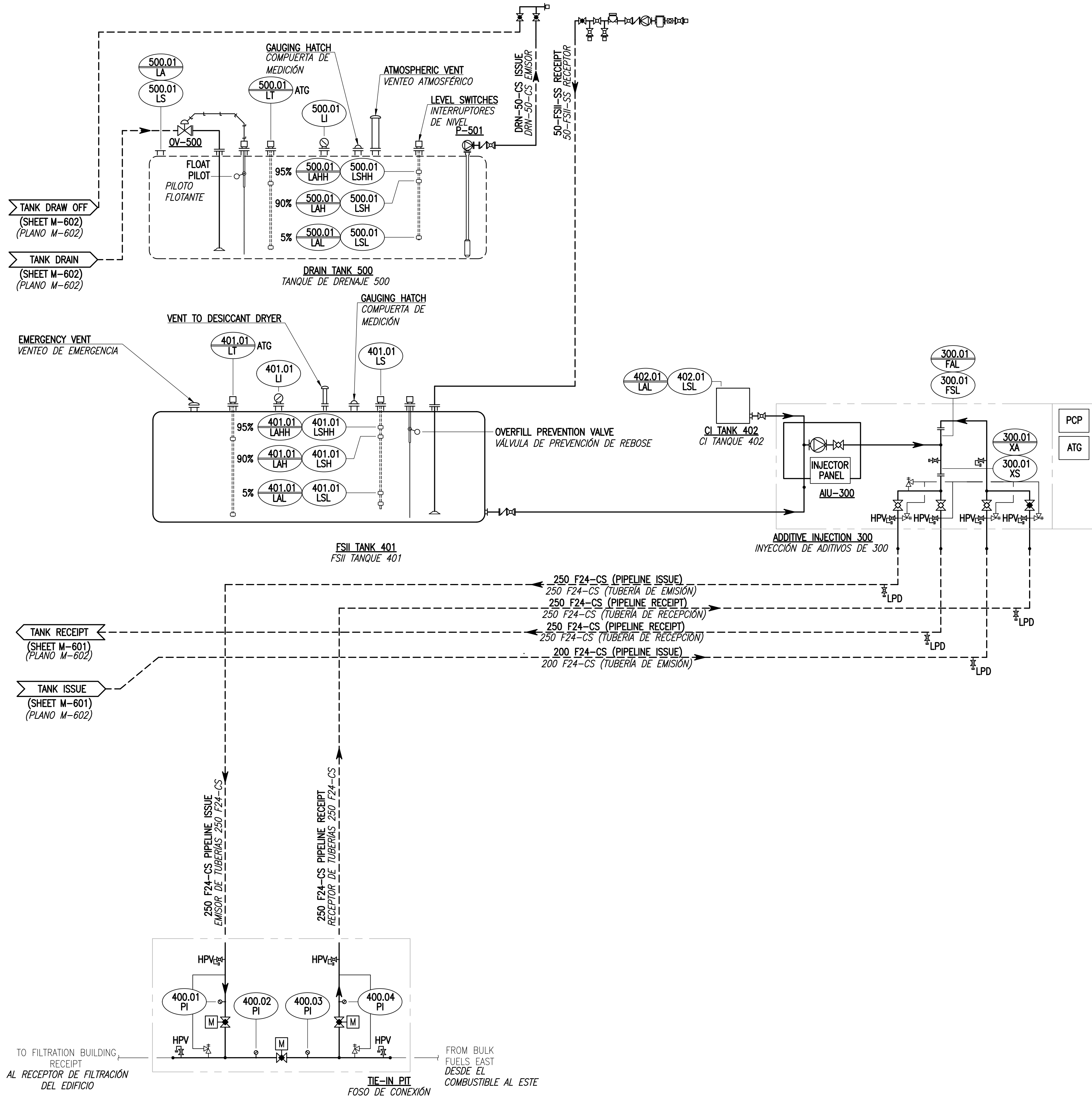
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PIPING & INSTRUMENT DIAGRAM (STORAGE TANKS)
DIAGRAMA DE TUBERÍAS E INSTRUMENTACIÓN (TANQUES DE ALMACENAJE)
SCALE: NONE
ESCALA: SIN ESCALA

APPR	
DATE	
SYN	
DESCRIPTION	
	
	
	
	
APPROVED	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO: DATE	
DES: BMS	DRW: RTH
FM/SM: CCA/CLA	CHK: DWN
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
ATLANTIC DESIGN AND CONSTRUCTION	
NAVAL STATION ROTA	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
PIPING & INSTRUMENT DIAGRAM (STORAGE TANKS)	
DIAGRAMA DE TUBERÍAS E INSTRUMENTACIÓN (TANQUES DE ALMACENAJE)	
SCALE:	
EPROJECT NO.:	
CONSTR. CONTR. NO.:	
NAVAFAC DRAWING NO. 14140191	
SHEET 87 OF 342	
M-601T	
NAVAFAC METRIC DRAWING REVISION: 07 AUGUST 2018	

FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\M-M-602.dwg LAYOUT NAME: m-602 PLOTTED: Tuesday, July 09, 2024 - 14:33pm USER: mhopkins



PIPING & INSTRUMENT DIAGRAM (ADDITIVE SYSTEM)
DIAGRAMA DE TUBERÍAS E INSTRUMENTACIÓN (SISTEMA DE ADITIVOS)
SCALE: NONE
ESCALA: SIN ESCALA

APPR	DATE	SYN	DESCRIPTION
APPROVED			
FOR COMMANDER NAVFAC			
ACTIVITY			
SATISFACTORY TO: DATE			
DES	BMS	DRW	RTH
FM/SM	CC/EA	CHK	DWN
BRANCH MANAGER			
DESIGN MANAGER			
FIRE PROTECTION			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC			
NAVAL STATION ROTA			
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1			
PIPING & INSTRUMENT DIAGRAM (ADDITIVE SYSTEM)			
DIAGRAMA DE TUBERÍAS E INSTRUMENTACIÓN (SISTEMA DE ADITIVOS)			
SCALE:			
EPROJECT NO.:			
CONSTR. CONTR. NO.:			
NAVFAC DRAWING NO. 14140192			
SHEET 88 OF 342			
M-602T			

[illegible]

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ADDITIVE INJECTION SYSTEM:

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FILE NAME: G:\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAD\NAVFAC\TANK VAULT HVAC PLAN.dwg LAYOUT NAME: mh401 PLOTTED: Tuesday, July 09, 2024 - 1:56pm USER: rhoplins

D

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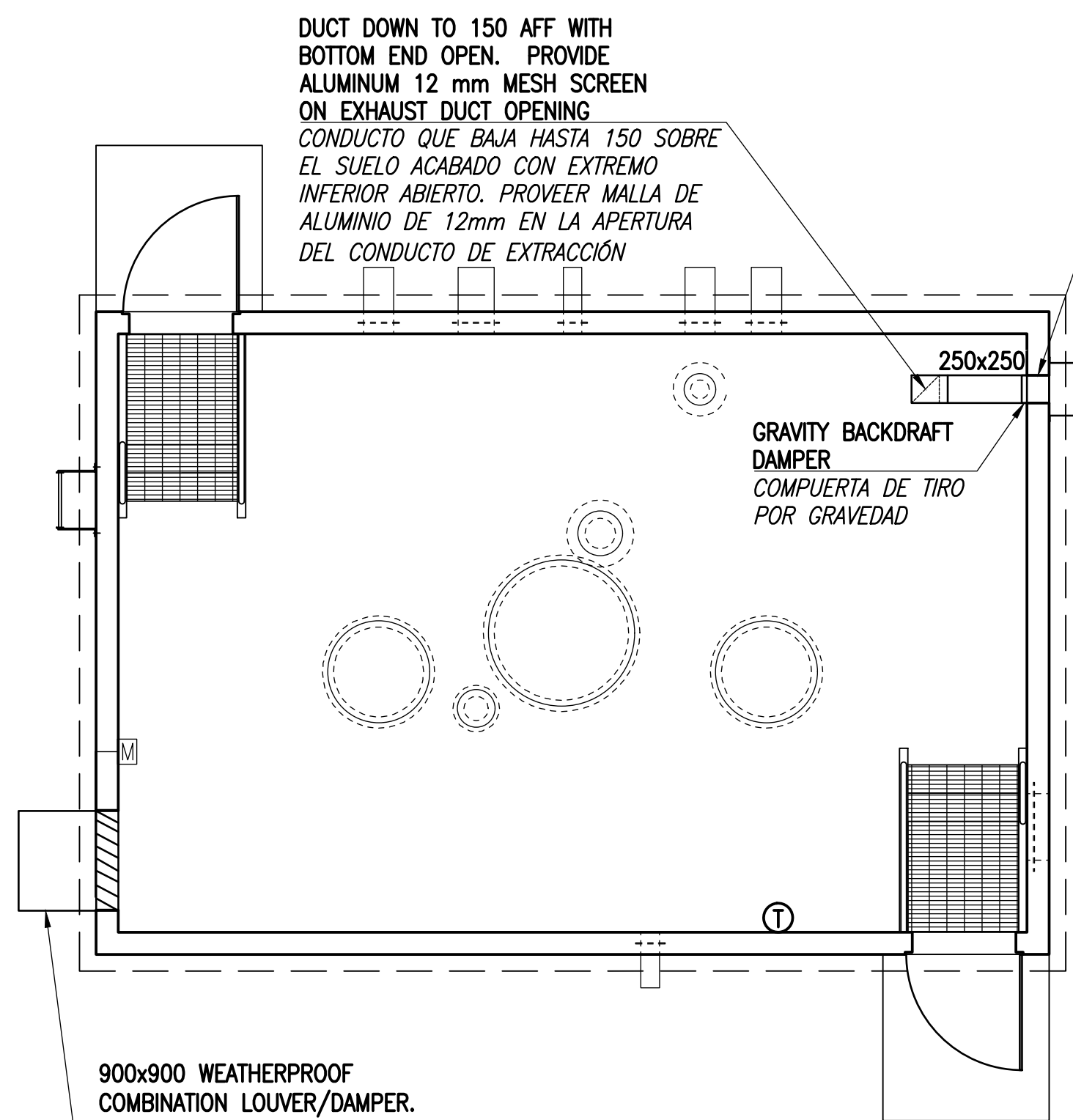
1

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PLAN NORTH
HVAC FLOOR PLAN - PUMP VAULT 100
PLANTA DE CLIMA - CÁMARA DE BOMBAS 100
SCALE: 1:50
ESCALA: 1:50

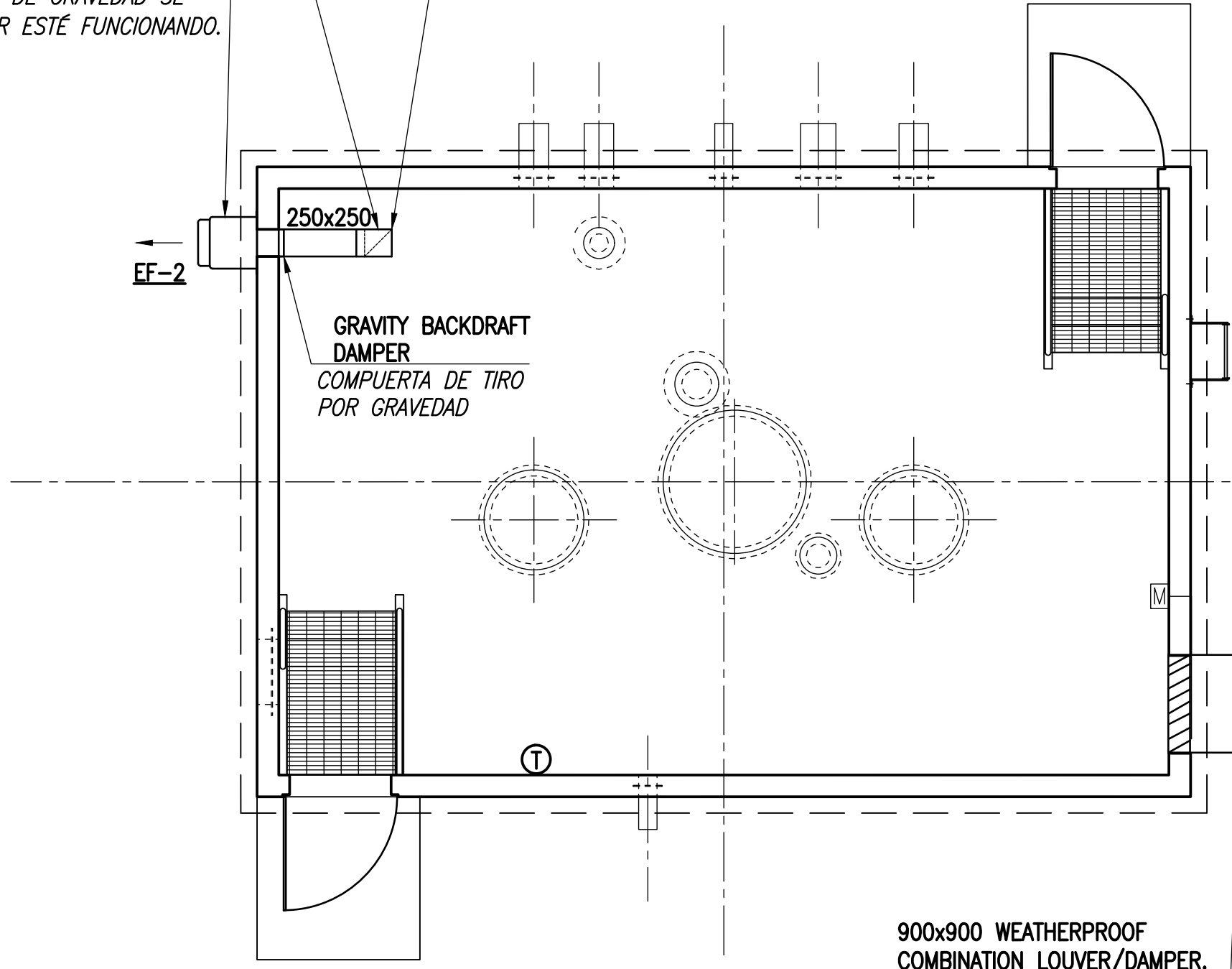
ABBREVIATIONS
ABREVIATURAS

A	AMPS	RA	RETURN AIR
///	///	///	///
AFF	ABOVE FINISHED FLOOR	L/S	LITERS PER SECOND
///	///	///	///
EER	ENERGY EFFICIENCY RATIO	PA	PASCAL
///	///	///	///
EF	EXHAUST FAN	PH	PHASE
///	///	///	///
EAT	ENTERING AIR TEMPERATURE	HZ	HERTZ
///	///	///	///
EXT SP	EXTERNAL STATIC PRESSURE	RPM	ROTATION
///	///	///	///
DB	DRY BULB	HVAC	VENTILATION, AND AIR CONDITIONING
///	///	///	///
MCWB	MEAN COINCIDENT WET BULB	IU	INTERNAL UNIT
///	///	///	///
MOCP	MAXIMUM OVERCURRENT PROTECTION	OU	OUTDOOR UNIT
///	///	///	///
KW	KILOWATT	SEER	SEASONAL ENERGY EFFICIENCY RATIO
///	///	///	///
HSPF	HEATING SEASONAL PERFORMANCE FACTOR	WB	WET BULB
///	///	///	///
HVAC	HEATING, VENTILATION, AND AIR CONDITIONING	V	VOLTS
///	///	///	///
CD	CONDENSATE		
///	///		
SA	SUPPLY AIR		
///	///		

915 MM STRAIGHT DUCT BEFORE ELBOW
CONDUCTO RECTO DE 915 mm ANTES DEL CODO.

FAN TO RUN WHEN ROOM TEMPERATURE IS ABOVE 26.7°C OR WHEN TURNED ON BY SWITCH. GRAVITY DAMPER TO OPEN WHEN FAN ON.
EL VENTILADOR ARRANCARÁ CUANDO LA TEMPERATURA DE LA HABITACIÓN SEA SUPERIOR A 26,7°C O CUANDO SE ENCIENDA CON EL INTERRUPTOR. LA COMPUERTA DE GRAVEDAD SE ABRIRÁ CUANDO EL VENTILADOR ESTÉ FUNCIONANDO.

EF-2



PLAN NORTH
HVAC FLOOR PLAN - PUMP VAULT 200
PLANTA DE CLIMA - CÁMARA DE BOMBAS 200
SCALE: 1:50
ESCALA: 1:50

HVAC NOTES:
NOTAS DE CLIMATIZACIÓN:

- THESE DRAWINGS ARE DIAGRAMMATIC IN NATURE AND ARE NOT INTENDED TO SHOW THE EXACT LOCATIONS OF COMPONENTS, NOR SHOW ALL SYSTEM COMPONENTS. CONTRACTOR MUST VERIFY STRUCTURAL AND ELECTRICAL INSTALLATIONS AND DRAWINGS TO AVOID OBSTRUCTIONS OR INTERFERENCE.
1. ESTOS PLANOS SON DE NATURALEZA ESQUEMÁTICA Y NO PRETENDEN MOSTRAR LA POSICIÓN EXACTA DE LOS COMPONENTES, NI MOSTRAR TODOS LOS COMPONENTES DEL SISTEMA. EL CONTRATISTA VERIFICARÁ LAS INSTALACIONES ESTRUCTURALES Y ELÉCTRICAS Y LOS PLANOS PARA EVITAR OBSTRUCCIONES O INTERFERENCIAS.
- MOUNT ALL MECHANICAL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.
2. MONTAR TODOS LOS EQUIPOS MECÁNICOS DE ACUERDO CON LAS INSTRUCCIONES DEL FABRICANTE.

GRAPHIC SCALE(S):

ESCALA(S) GRÁFICA(S):



APPR	DATE	SYN	DESCRIPTION
APPROVED			
FOR COMMANDER NAVFAC			
ACTIVITY			
SATISFACTORY TO: DATE			
DES	DWN	DRW	RTH CHK CSB
FM/DM	CCA/EAL		
BRANCH MANAGER			
DESIGN MANAGER			
FIRE PROTECTION			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC			
NAVAL STATION ROTA			
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1			
TANK VAULT HVAC PLAN			
PLANTA DE CLIMATIZACIÓN DE LA CÁMARA DEL TANQUE			
SCALE:			
EPROJECT NO.:			
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO. 14140196			
SHEET 92 OF 342			
MH401T			

C:\Users\thopkins\Documents\21-015 BLDG_AIB_CENTRAL MODEL_rhopkins@brockenbrough.com.rvt

D

C

B

A

HVAC NOTES NOTAS DE CLIMATIZACIÓN

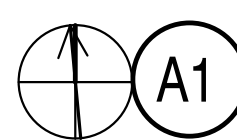
1. THESE DRAWINGS ARE DIAGRAMMATIC IN NATURE AND ARE NOT INTENDED TO SHOW THE EXACT LOCATIONS OF COMPONENTS, NOR SHOW ALL SYSTEM COMPONENTS. CONTRACTOR MUST VERIFY STRUCTURAL AND ELECTRICAL INSTALLATIONS AND DRAWINGS TO AVOID OBSTRUCTIONS OR INTERFERENCE.

ESTOS PLANOS TIENEN CARÁCTER ESQUEMÁTICO Y NO SE ENTIENDEN COMO QUE MUESTRAN LAS POSICIONES EXACTAS DE LOS COMPONENTES, NI MUESTRAN TODOS LOS COMPONENTES DEL SISTEMA EL CONTRATISTA DEBERÁ COMPROBAR LAS INSTALACIONES Y PLANOS DE ESTRUCTURA Y ELECTRICIDAD PARA EVITAR OBSTRUCCIONES O INTERFERENCIAS.

2. MOUNT ALL MECHANICAL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.

MONTAR TODOS LOS EQUIPOS MECÁNICOS DE ACUERDO CON LAS INSTRUCCIONES DEL FABRICANTE.

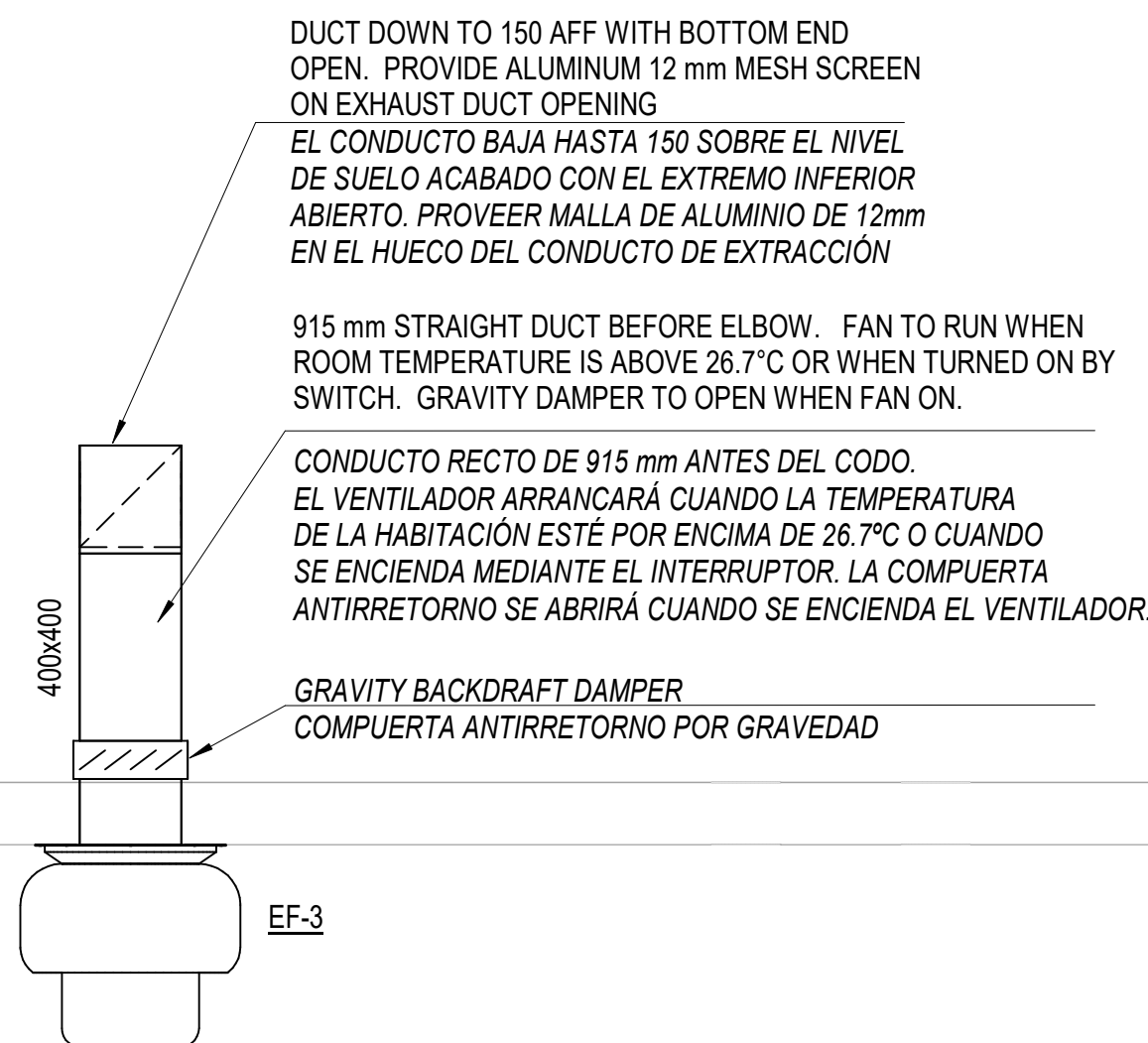
PLAN
NORTH



HVAC PLAN - INJECTOR BUILDING

PLANTA DE CLIMATIZACIÓN – EDIFICIO DEL INYECTOR

SCALE: 1" = 30'
ESCALA: 1" = 30'

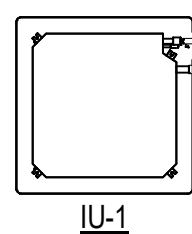


EQUIPMENT ROOM
300

1200x1350 WEATHERPROOF COMBINATION LOUVER/DAMPER. INTERLOCK WITH EF-3.
COMBINACIÓN DE REJILLA / COMPUERTA DE 1200x1350 RESISTENTES A LA INTemperie. INTERCONECTAR CON EF-3.

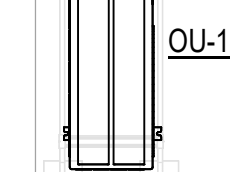
1200x1350 DUCT SLEEVE PASATUBOS PARA CONDUCTO DE 1200x1350

ELECTRICAL ROOM
301

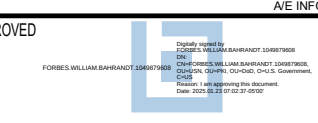
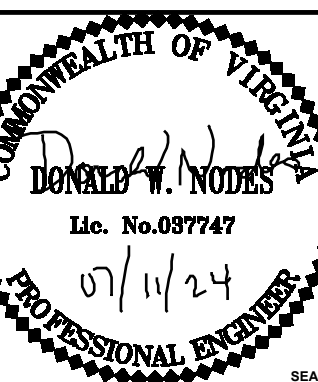


32mm CD ROUTE TO SPLASH BLOCK OUTSIDE BUILDING
CONDUCIR DRENAJE DE CONDENSADO DE 32mm HASTA SALPICADERO EXTERIOR DEL

PRE-INSULATED REFRIGERANT LINE SET
CONJUNTO DE LÍNEAS DE REFRIGERANTE PREAisladas



GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):



SATISFACTORY TO: DATE: DES DWN DRW DKW CHK CSB
PACOM: CCA/EIL
BRANCH MANAGER:
DESIGN MANAGER:

NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ ATLANTIC
ATLANTIC DESIGN AND CONSTRUCTION
NAVAL STATION ROTA
ROTA, SPAIN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
HVAC PLAN - INJECTOR BUILDING
PLANTA DE CLIMATIZACIÓN – EDIFICIO DEL INYECTOR

SCALE: AS NOTED
PROJECT NO.:
CONSTR. CONTR. NO.:

NAVFAC DRAWING NO.
14140197
SHEET 93 OF 342

MH402T

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

NAVFAC

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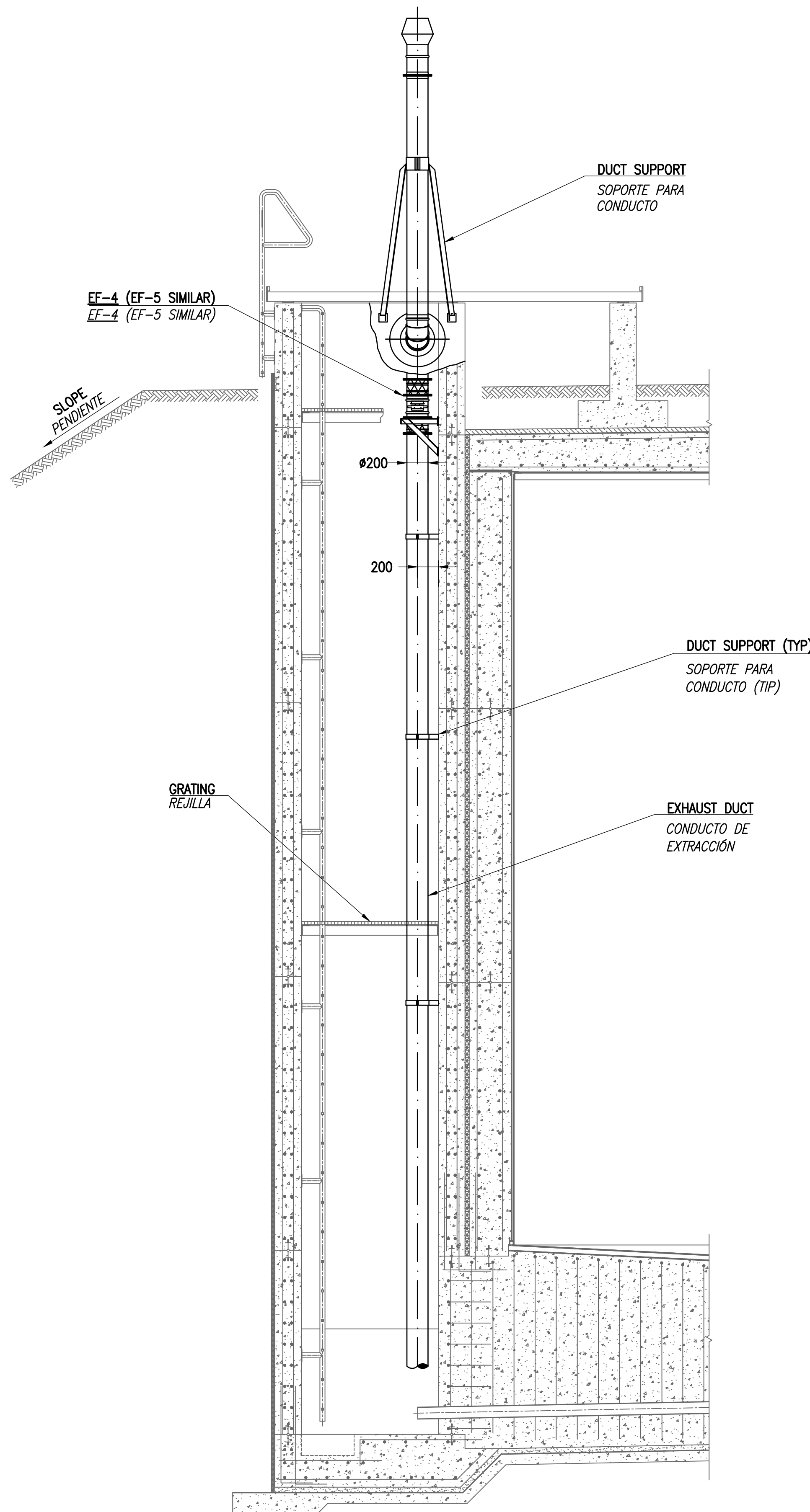
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D

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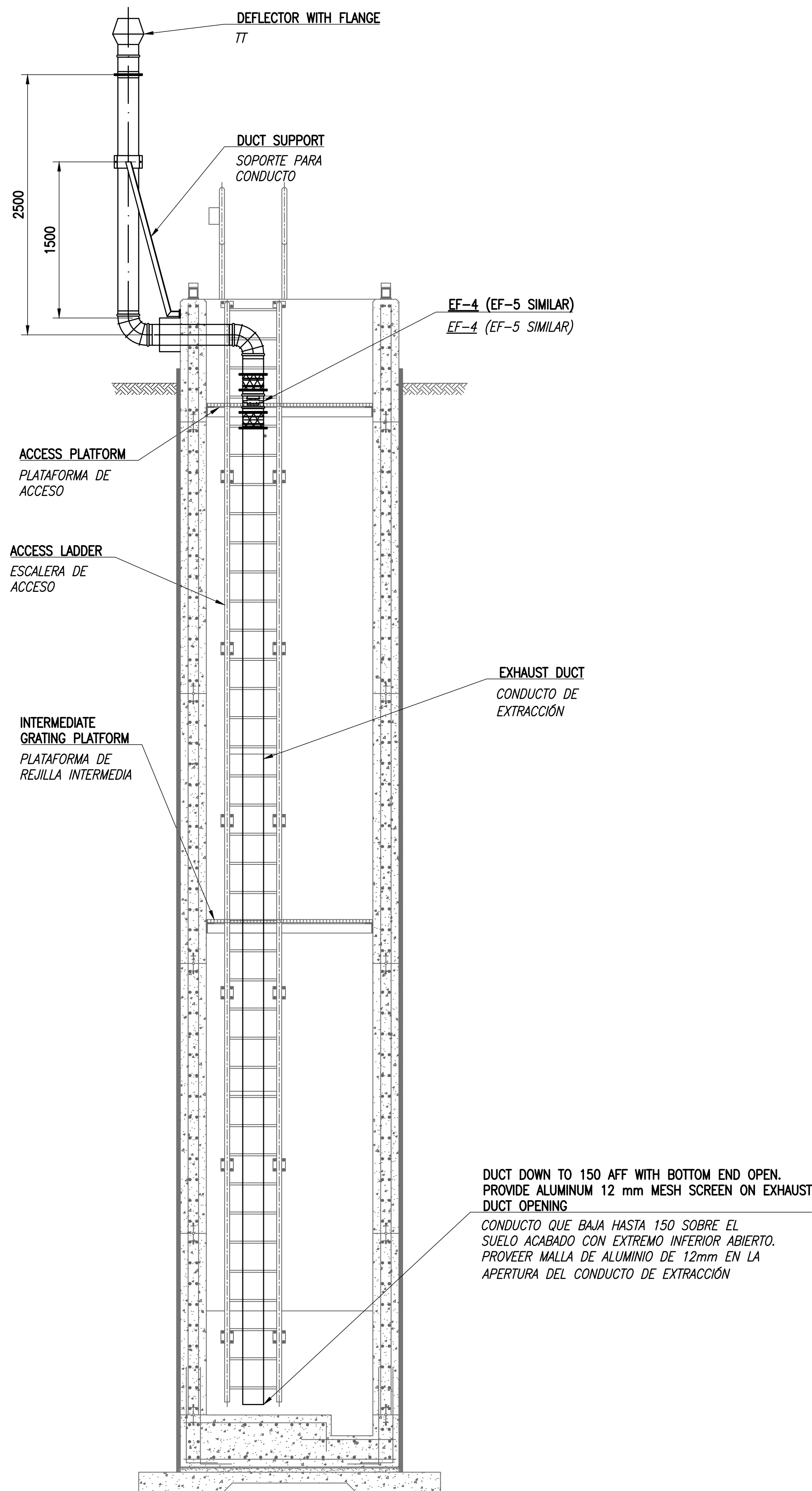
B

A



A1

LEAK DETECTION WELL SECTION 1
POZO DE DETECCIÓN DE FUGAS SECCIÓN 1
SCALE: 1:30
ESCALA: 1:30



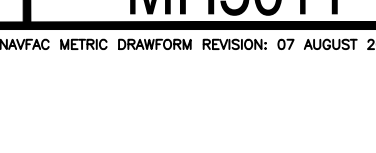
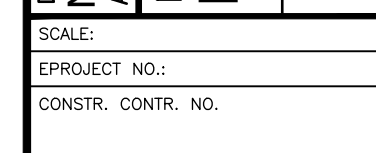
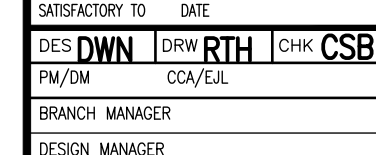
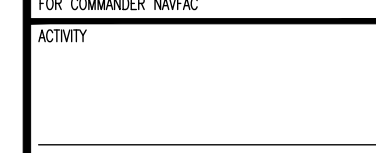
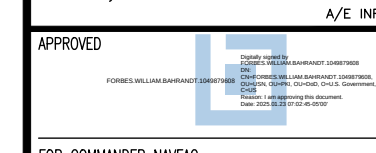
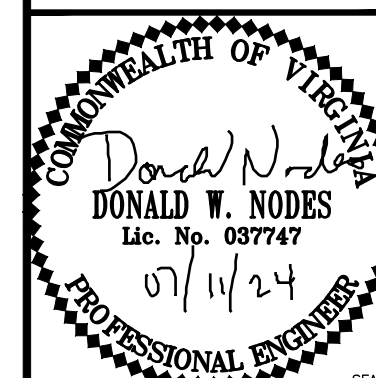
A3

LEAK DETECTION WELL SECTION 2
POZO DE DETECCIÓN DE FUGAS SECCIÓN 2
SCALE: 1:30
ESCALA: 1:30

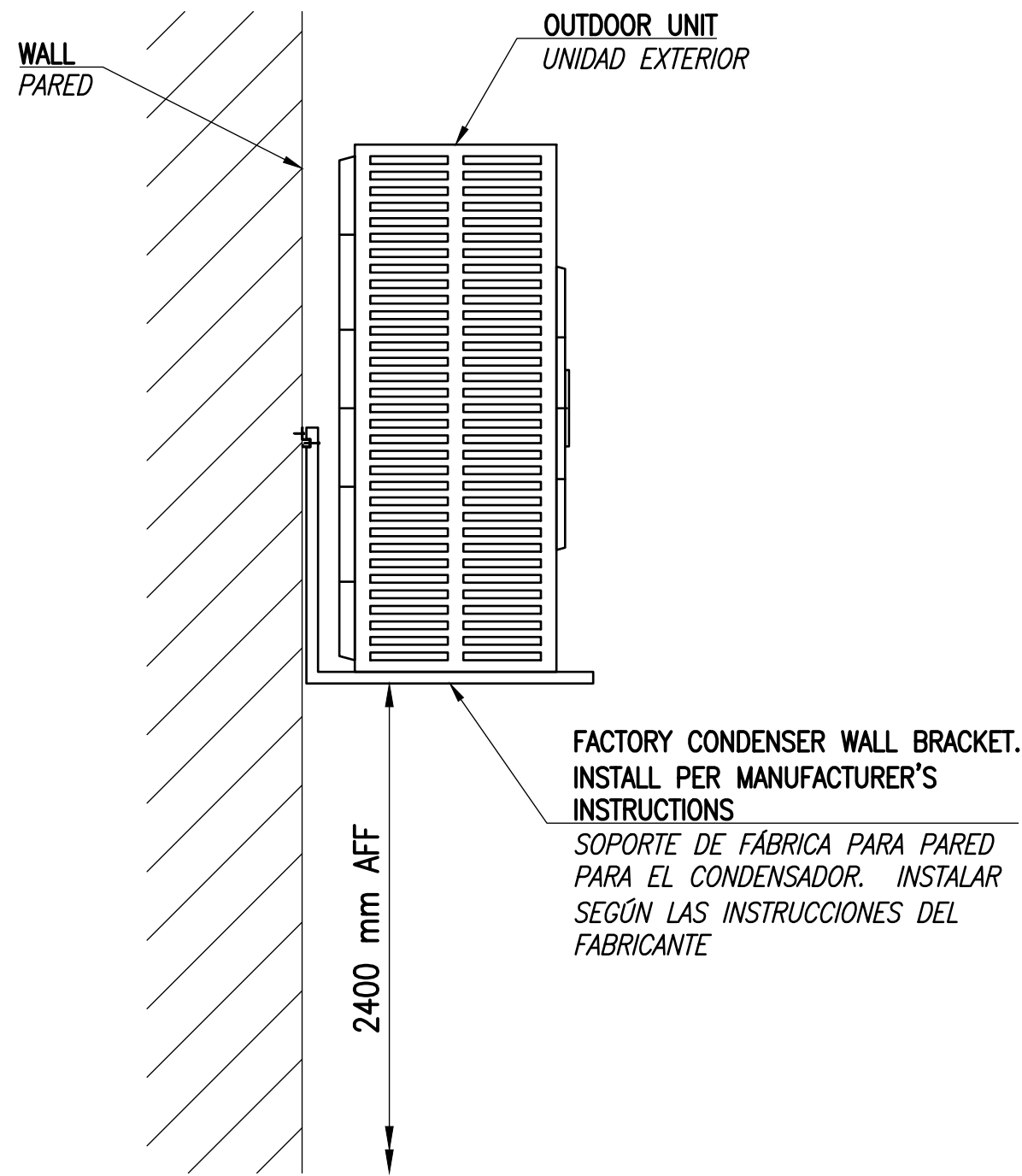
NOTE:
EF-4 AND EF-5 TO BE CONTROLLED
BY A LOCAL START/STOP SWITCH.
TT

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):

1:30 0 1000 2000 3000 mm

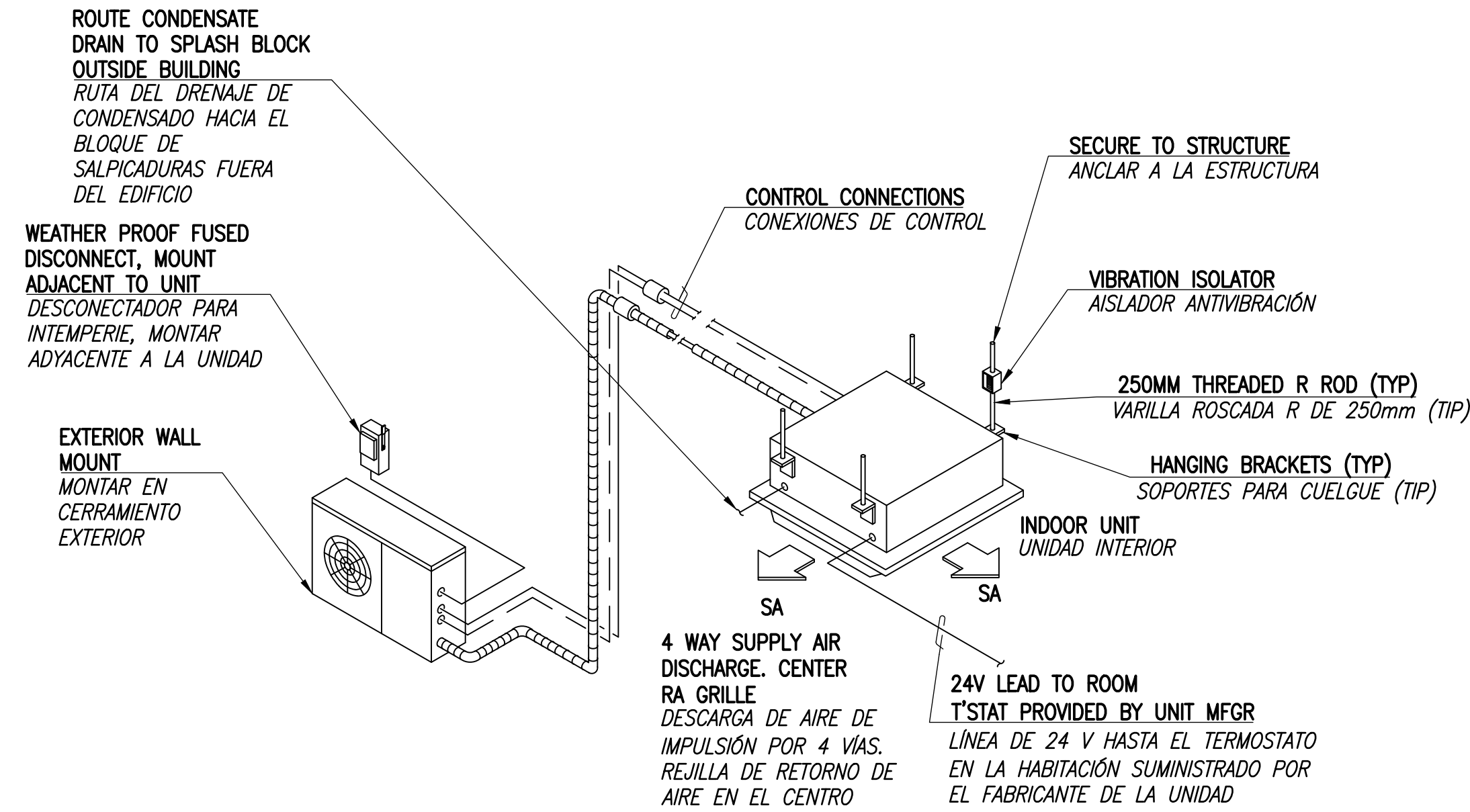


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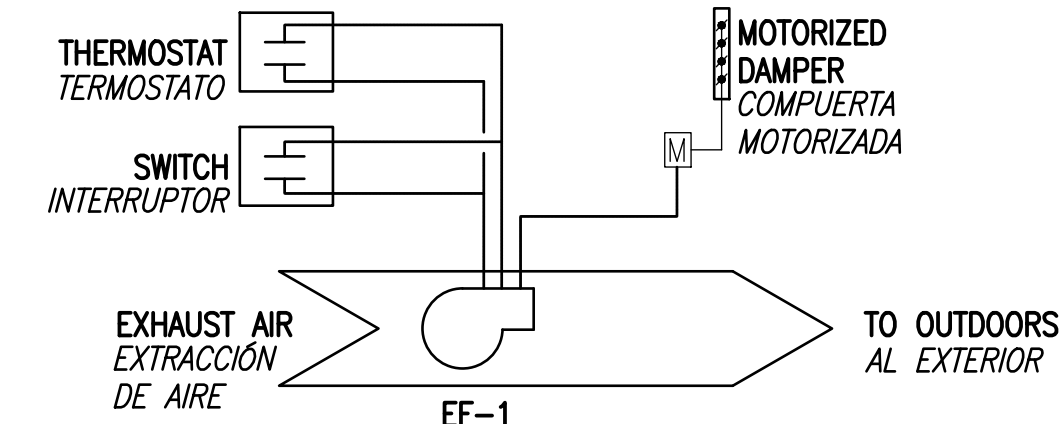
OUTDOOR UNIT WALL MOUNTING DETAIL
DETALLE DE MONTAJE EN PARED DE UNIDAD EXTERIOR

SCALE: 1:10
ESCALA: 1:10



DUCTLESS AIR CONDITIONER UNIT
UNIDAD DE AIRE ACONDICIONADO SIN CONDUCTOS

SCALE: NONE
ESCALA: SIN ESCALA



NOTE: TYPICAL OF EF-2 AND EF-3
NOTA: TÍPICO PARA EF-2 Y EF-3

EF-1 CONTROL SCHEMATIC
ESQUEMA DE CONTROL EF-1

SCALE: NONE
ESCALA: SIN ESCALA

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):

1:10 0 100 200 500 1000 mm

APPROVED	DATE	DESCRIPTION	NAME
APPROVED			
FOR COMMANDER NAVFAC			
ACTIVITY			
SATISFACTORY TO: DATE			
DES: DWN	DRW: RTH	CHK: CSB	
PM/SM	CC/EA		
BRANCH MANAGER			
DESIGN MANAGER			
FIRE PROTECTION			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC			
NAVAL STATION ROTA			
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1			
DETAILS			
SCALE:			
EPROJECT NO.:			
CONSTR. CONTR. NO.:			
NAVFAC DRAWING NO. 14140199			
SHEET 95 OF 342			
MH502T			

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

NAVJAG METRIC DRAWFORM REVISION: 07 AUGUST 2018